MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 1 SOLUTIONS

1. Prove that the following are vector spaces over $\mathbb{R}$:
(a) polynomials of degree not more than d,

$$\mathbb{P}_d = \{a_0 + a_1 x + \cdots + a_d x^d \mid a_i \in \mathbb{R} \text{ for all } i\},$$

(b) m-by-n matrices

$$\mathbb{R}^{m\times n} = \{[a_{ij}]_{i,j=1}^{m,n} \mid a_{ij} \in \mathbb{R} \text{ for all } i, j\}.$$

The addition and scalar multiplication operations for polynomials and matrices are as defined in the lectures.
SOLUTION. Routine.

2. In class, we have shown that $\mathbb{R}_+ = \{x \in \mathbb{R} \mid x > 0\}$ is a vector space over $\mathbb{R}$ with addition and scalar multiplication defined by

$$x \oplus y = xy \quad \text{and} \quad \lambda \odot x = x^\lambda$$

for $x, y \in \mathbb{R}_+$, $\lambda \in \mathbb{R}$. Is the statement still true if we change the definition of scalar multiplication to

$$\lambda \odot x = \lambda^x$$

instead? If it is true, prove it. Otherwise, show why it is not true.
SOLUTION. $\mathbb{R}_+$ is not a vector space over $\mathbb{R}$ with respect to scalar multiplication defined by $\lambda \odot x = \lambda^x$. For example, $2 \odot (1 \oplus 1) = 2$ but $(2 \odot 1) \oplus (2 \odot 1) = 4$ and thus the distributive law for scalar multiplication over vector addition is not satisfied.

3. Let V be a vector space over $\mathbb{R}$. Suppose V has only a finite number of elements. What can we deduce about V?
SOLUTION. V has to be the zero vector space. Clearly $V = \{\mathbf{0}\}$ is a vector space over $\mathbb{R}$ containing a finite number (ie. one) of elements. Suppose V contains a non-zero element $\mathbf{v}$. Then since a vector space must be closed under scalar multiplication, V must also contain $\alpha \cdot \mathbf{v}$ for all $\alpha \in \mathbb{R}$. But given any $\alpha, \beta \in \mathbb{R}$ where $\alpha \neq \beta$, we must have that

$$\alpha \cdot \mathbf{v} \neq \beta \cdot \mathbf{v};$$

otherwise, $\alpha \cdot \mathbf{v} = \beta \cdot \mathbf{v}$ implies $(\alpha - \beta) \cdot \mathbf{v} = \mathbf{0}$ (by Problem **4**(d)) which in turn implies that $\alpha = \beta$ (by Theorem **1.3**(d) and the fact that $\mathbf{v} \neq \mathbf{0}$). In other words, every $\alpha \cdot \mathbf{v}$ is distinct for distinct $\alpha \in \mathbb{R}$. Since $\mathbb{R}$ is infinite, so is the set

$$\{\alpha \cdot \mathbf{v} \mid \alpha \in \mathbb{R}\} \subseteq V$$

and so is V. Hence we have shown that any V that contains a non-zero vector must be infinite — which means that the only V that is finite is the zero vector space.

4. Let V be a vector space over a field $\mathbb{F}$. The additive identity (a.k.a. zero vector) of V is denoted by $\mathbf{0}$. The additive inverse (a.k.a. negative vector) of a vector $\mathbf{v} \in V$ is denoted by $-\mathbf{v}$.

Date: September 23, 2007 (Version 1.0).

小红书
小红书号:3189205503 qq

(a) Prove that

$$\lambda \cdot \mathbf{0} = \mathbf{0}$$

for all $\lambda \in \mathbb{F}$.

SOLUTION. Note that

$$\lambda \cdot \mathbf{0} + \lambda \cdot \mathbf{0} = \lambda \cdot (\mathbf{0} + \mathbf{0}) = \lambda \cdot \mathbf{0}$$

where the first equality is by the distributivity of scalar multiplication over vector addition and the second equality is by the property of the zero vector/additive identity. Now by the existence of negative vector/additive inverse, we know that there is an element $\mathbf{v} \in V$ such that

$$\mathbf{v} + \lambda \cdot \mathbf{0} = \mathbf{0}. \tag{4.1}$$

Adding $\mathbf{v}$ to both sides of $\lambda \cdot \mathbf{0} + \lambda \cdot \mathbf{0} = \lambda \cdot \mathbf{0}$ yields

$$\mathbf{v} + (\lambda \cdot \mathbf{0} + \lambda \cdot \mathbf{0}) = \mathbf{v} + \lambda \cdot \mathbf{0}. \tag{4.2}$$

The LHS of (4.2) may be simplied via

$$\mathbf{v} + (\lambda \cdot \mathbf{0} + \lambda \cdot \mathbf{0}) = (\mathbf{v} + \lambda \cdot \mathbf{0}) + \lambda \cdot \mathbf{0} = \mathbf{0} + \lambda \cdot \mathbf{0} = \lambda \cdot \mathbf{0} \tag{4.3}$$

where the equalities are due to the associativity of vector addition, the property of the negative vector/additive inverse, and the property of zero vector/additive identity. Substituing (4.1) and (4.3) into (4.2) yields

$$\lambda \cdot \mathbf{0} = \mathbf{0}$$

as required.

(b) Prove that

$$-(-\mathbf{v}) = \mathbf{v}$$

for all $\mathbf{v} \in V$.

SOLUTION. Note that

$$(-\mathbf{v}) + \mathbf{v} = 0 = \mathbf{v} + (-\mathbf{v})$$

by the definition of $-\mathbf{v}$. But this same equation also tells us that the additive inverse/negative vector of $-\mathbf{v}$, ie. $-(-\mathbf{v})$, is $\mathbf{v}$.

(c) Let the additive identity of $\mathbb{F}$ be denoted by 0 and the additive inverse of $\lambda \in \mathbb{F}$ be denoted by λ' (thus $\lambda + \lambda' = 0 = \lambda' + \lambda$ in $\mathbb{F}$). Prove that

$$\lambda \cdot (-\mathbf{v}) = -(\lambda \cdot \mathbf{v}) = \lambda' \cdot \mathbf{v}$$

for all $\lambda \in \mathbb{F}$ and all $\mathbf{v} \in V$. Note that there are two equalities to be proved.

SOLUTION. First we will prove that

$$-(\lambda \cdot \mathbf{v}) = \lambda' \cdot \mathbf{v}.$$

Note that

$$\mathbf{0} = 0 \cdot \mathbf{v} = (\lambda + \lambda') \cdot \mathbf{v} = \lambda \cdot \mathbf{v} + \lambda' \cdot \mathbf{v}$$

where the first equality is by Theorem **1.3**(a), the second by the property of λ' as the additive inverse of $\lambda \in \mathbb{F}$, and the third by distributivity of scalar multiplication over scalar addition. Together with the commutativity of vector addition, we obtain

$$\lambda \cdot \mathbf{v} + \lambda' \cdot \mathbf{v} = \mathbf{0} = \lambda' \cdot \mathbf{v} + \lambda \cdot \mathbf{v}$$

which means that $\lambda' \cdot \mathbf{v}$ is the negative vector/additive inverse of $\lambda \cdot \mathbf{v}$, ie.

$$\lambda' \cdot \mathbf{v} = -(\lambda \cdot \mathbf{v}).$$

Next we will prove that

$$\lambda \cdot (-\mathbf{v}) = -(\lambda \cdot \mathbf{v}).$$

Note that

$$\mathbf{0} = \lambda \cdot \mathbf{0} = \lambda \cdot (-\mathbf{v} + \mathbf{v}) = \lambda \cdot (-\mathbf{v}) + \lambda \cdot \mathbf{v}$$

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小红书号:3189205503 qq

where the first equality is by Problem **4**(a), the second by the property of negative vector/additive inverse, the third by the distributivity of scalar multiplication over vector addition. Together with the commutativity of vector addition, we obtain

$$\lambda \cdot \mathbf{v} + \lambda \cdot (-\mathbf{v}) = \mathbf{0} = \lambda \cdot (-\mathbf{v}) + \lambda \cdot \mathbf{v}$$

which means that $\lambda \cdot (-\mathbf{v})$ is the negative vector/additive inverse of $\lambda \cdot \mathbf{v}$, ie.

$$\lambda \cdot (-\mathbf{v}) = -(\lambda \cdot \mathbf{v}).$$

(d) Recall that subtracting a vector $\mathbf{v}$ from a vector $\mathbf{u}$ is defined as adding the additive inverse of $\mathbf{v}$ to $\mathbf{u}$, ie. $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$. Prove that

$$\lambda \cdot (\mathbf{u} - \mathbf{v}) = \lambda \cdot \mathbf{u} - \lambda \cdot \mathbf{v}$$

and

$$(\lambda - \mu) \cdot \mathbf{u} = \lambda \cdot \mathbf{u} - \mu \cdot \mathbf{u}$$

for all $\mathbf{u}, \mathbf{v} \in V$, $\lambda \in \mathbb{F}$.
SOLUTION. Note that

$$\begin{aligned} \lambda \cdot \mathbf{u} - \lambda \cdot \mathbf{v} &= \lambda \cdot \mathbf{u} + [-(\lambda \cdot \mathbf{v})] \\ &= \lambda \cdot \mathbf{u} + \lambda \cdot (-\mathbf{v}) \\ &= \lambda \cdot [\mathbf{u} + (-\mathbf{v})] \\ &= \lambda \cdot [\mathbf{u} - \mathbf{v}] \end{aligned}$$

where the first and last equalities are by the definition of vector subtraction, the second equality is by Problem **4**(c), and the third equality is by the distributivity of scalar multiplication over vector addition.
For the second identity, note that

$$\begin{aligned} \lambda \cdot \mathbf{u} - \mu \cdot \mathbf{u} &= \lambda \cdot \mathbf{u} + [-(\mu \cdot \mathbf{u})] \\ &= \lambda \cdot \mathbf{u} + \mu' \cdot \mathbf{u} \\ &= (\lambda + \mu') \cdot \mathbf{u} \end{aligned}$$

where the first equality is by the definition of vector subtraction, the second equality is by Problem **4**(c), and the third equality is by the distributivity of scalar multiplication over scalar addition. Now just observe that $-\mu$ is just another (more common) way of writing μ', the additive inverse of μ in $\mathbb{F}$.

5. Let V be a vector space over $\mathbb{R}$ with addition and scalar multiplication denoted by $+$ and $\cdot$ respectively. Let $W = V \times V = \{(\mathbf{v}_1, \mathbf{v}_2) \mid \mathbf{v}_1, \mathbf{v}_2 \in V\}$. Prove that W is a vector space over $\mathbb{C}$ with addition defined by

$$(\mathbf{u}_1, \mathbf{u}_2) \boxplus (\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{v}_1, \mathbf{u}_2 + \mathbf{v}_2)$$

for all $(\mathbf{u}_1, \mathbf{u}_2), (\mathbf{v}_1, \mathbf{v}_2) \in W$ and scalar multiplication defined by

$$(a + bi) \boxdot (\mathbf{v}_1, \mathbf{v}_2) = (a \cdot \mathbf{v}_1 - b \cdot \mathbf{v}_2, b \cdot \mathbf{v}_1 + a \cdot \mathbf{v}_2)$$

for all $a + bi \in \mathbb{C}$ and $(\mathbf{v}_1, \mathbf{v}_2) \in W$. Here $i = \sqrt{-1}$ and $a, b \in \mathbb{R}$.
SOLUTION. Routine.

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MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 2 SOLUTIONS

1. Let $\mathbb{P}_7$ be the vector space of polynomials of degree not more than 7, ie.

$$\mathbb{P}_7 = \{p(x) = a_0 + a_1x + \cdots + a_7x^7 \mid a_i \in \mathbb{R} \text{ for all } i\}.$$

Consider the following subsets of $\mathbb{P}_7$:
(a) $W_a = \{p(x) \in \mathbb{P}_7 \mid \text{degree of } p(x) \text{ is } 3\}$,
(b) $W_b = \{p(x) \in \mathbb{P}_7 \mid \text{degree of } p(x) \text{ is not more than } 3\}$,
(c) $W_c = \{p(x) \in \mathbb{P}_7 \mid 2p(0) = p(1)\}$,
(d) $W_d = \{p(x) \in \mathbb{P}_7 \mid p(x) \geq 0 \text{ for all } x \text{ with } 0 \leq x \leq 1\}$,
(e) $W_e = \{p(x) \in \mathbb{P}_7 \mid p(x) = p(1-x) \text{ for all } x\}$.
Which of these are subspaces of $\mathbb{P}_7$? Justify your answer.
SOLUTION. W_a is not a subspace since the zero polynomial does not have degree 3 and is thus not in W_a. W_b is a subspace — it is clearly not empty; for any $p, q \in W_b$, $\deg(p), \deg(q) \leq 3$ and so $\deg(\lambda p + \mu q) \leq 3$, and so $\alpha p + \beta q \in W_b$. W_d is not a subspace since it is not closed under scalar multiplication, for example, $x^2 \in W_d$ but $-1 \cdot x^2 \notin W_d$. W_c and W_e are both subspaces — they clearly contain the zero polynomial and are thus nonempty; it is straightforward to check that if p, q satisfy the condition of being in W_c or W_e, then so would $\lambda p + \mu q$.

2. Let W_1 and W_2 be subspaces of a vector space V. Suppose W_1 is neither the zero subspace $\{\mathbf{0}\}$ nor the vector space V itself and likewise for W_2. Show that there exists a vector $\mathbf{v} \in V$ such that $\mathbf{v} \notin W_1$ and $\mathbf{v} \notin W_2$. [If a subspace $W = \{\mathbf{0}\}$ or V, we call it a *trivial subspace* and otherwise we call it a *non-trivial subspace*.]
SOLUTION. If $W_1 \subseteq W_2$, then since $W_2 \neq V$, there must be a $\mathbf{v} \in V$ such that $\mathbf{v} \notin W_2$ (and thus $\mathbf{v} \notin W_1$). Likewise if $W_2 \subseteq W_1$. If $W_1 \nsubseteq W_2$ and $W_2 \nsubseteq W_1$, then there must eixst

$$\mathbf{v}_1 \in W_1, \quad \mathbf{v}_1 \notin W_2,$$

and

$$\mathbf{v}_2 \notin W_1, \quad \mathbf{v}_2 \in W_2.$$

Let $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$. Then

$$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 \notin W_1 \cup W_2.$$

Why? Because if not, suppose $\mathbf{v} \in W_1$, then W_1 is a subspace implies that $\mathbf{v}_2 = \mathbf{v} - \mathbf{v}_1 \in W_1$ — a contradiction (likewise if $\mathbf{v} \in W_2$). Hence $\mathbf{v} \notin W_1$ and $\mathbf{v} \notin W_2$.

3. Let W_1 and W_2 be subspaces of a vector space V. Let $W_1 + W_2$ be the subset of V defined by

$$W_1 + W_2 = \{\mathbf{w}_1 + \mathbf{w}_2 \in V \mid \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2\}.$$

(a) Prove that $W_1 + W_2$ is a subspace of V. [This subspace is called the *sum* of W_1 and W_2.]
SOLUTION. Note that $\mathbf{0} = \mathbf{0} + \mathbf{0} \in W_1 + W_2$ and so it is nonempty. Let $\alpha, \beta \in \mathbb{F}$. Let $\mathbf{w}_1 + \mathbf{w}_2, \mathbf{w}_1' + \mathbf{w}_2' \in W_1 + W_2$ where $\mathbf{w}_1, \mathbf{w}_1' \in W_1, \mathbf{w}_2, \mathbf{w}_2' \in W_2$. Since W_1 and W_2 are subspaces, $\alpha\mathbf{w}_1 + \beta\mathbf{w}_1' \in W_1$ and $\alpha\mathbf{w}_2 + \beta\mathbf{w}_2' \in W_2$. Hence

$$\alpha(\mathbf{w}_1 + \mathbf{w}_2) + \beta(\mathbf{w}_1' + \mathbf{w}_2') = (\alpha\mathbf{w}_1 + \beta\mathbf{w}_1') + (\alpha\mathbf{w}_2 + \beta\mathbf{w}_2') \in W_1 + W_2$$

and $W_1 + W_2$ is a subspace by Theorem 1.8.

Date: September 23, 2007 (Version 1.0).

小红书
小红书号:3189205503 qq

(b) Prove that $W_1 \cup W_2 \subseteq W_1 + W_2$.
SOLUTION. Let $\mathbf{w}_1 \in W_1$. Note that $\mathbf{0} \in W_2$ since it is a subspace. So $\mathbf{w}_1 = \mathbf{w}_1 + \mathbf{0} \in W_1 + W_2$. Therefore $W_1 \subseteq W_1 + W_2$. Likewise, $W_2 \subseteq W_1 + W_2$. Hence $W_1 \cup W_2 \subseteq W_1 + W_2$.

(c) Prove that $W_1 + W_2$ is the smallest subspace of V that contains $W_1 \cup W_2$. In other words, show that if U is any subspace of V such that

$$W_1 \cup W_2 \subseteq U,$$

then

$$W_1 + W_2 \subseteq U.$$

SOLUTION. Let $\mathbf{w}_1 + \mathbf{w}_2 \in W_1 + W_2$ where $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. Since $W_1 \cup W_2 \subseteq U$, we must have that $\mathbf{w}_1 \in U$ and $\mathbf{w}_2 \in U$. And since U is a subspace, we must also have that $\mathbf{w}_1 + \mathbf{w}_2 \in U$. Hence $W_1 + W_2 \subseteq U$.

(d) What is $W + W$?
SOLUTION. Take any $\mathbf{w} \in W$. Note that $\mathbf{0} \in W$ since it is a subspace. So $\mathbf{w} = \mathbf{w} + \mathbf{0} \in W + W$. Therefore $W \subseteq W + W$. Take any $\mathbf{w}_1 + \mathbf{w}_2 \in W + W$ where $\mathbf{w}_1 \in W$ and $\mathbf{w}_2 \in W$. Since W is a subspace, $\mathbf{w}_1 + \mathbf{w}_2 \in W$ by additive closure. Therefore $W + W \subseteq W$. It then follows that $W + W = W$.

(e) Under what condition is $W_1 \cup W_2$ a subspace? The condition I am looking for is not $W_1 \cup W_2 = W_1 + W_2$ (although this is true).
SOLUTION. $W_1 \cup W_2$ is a subspace if and only if either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$ and so is a subspace. If $W_2 \subseteq W_1$, then $W_1 \cup W_2 = W_1$ and so is a subspace. If $W_1 \nsubseteq W_2$ and $W_2 \nsubseteq W_1$, then $W_1 \cup W_2$ cannot be a subspace by the same argument in our solution of Problem **2**.

(f) Let W be any subspace of V. We know from basic set theory that

$$W \cap (W_1 \cup W_2) = (W \cap W_1) \cup (W \cap W_2).$$

But is the following true?

$$W \cap (W_1 + W_2) = (W \cap W_1) + (W \cap W_2)$$

SOLUTION. It is false. Here is a counter-example. Let $W = \{(x, y) \in \mathbb{R}^2 \mid x = y\}$, $W_1 = \{(x, y) \in \mathbb{R}^2 \mid y = 0\}$, $W_2 = \{(x, y) \in \mathbb{R}^2 \mid x = 0\}$. Then $W_1 + W_2 = \mathbb{R}^2$ since every $(x, y) \in \mathbb{R}^2$ may be written as $(x, y) = (x, 0) + (0, y) \in W_1 + W_2$. So

$$W \cap (W_1 + W_2) = W.$$

However, $W \cap W_1 = \{(0, 0)\} = W \cap W_2$ and so

$$(W \cap W_1) + (W \cap W_2) = \{(0, 0)\}.$$

(g) Show that if $W_1 \cap W_2 = \{\mathbf{0}\}$, then every $\mathbf{w} \in W_1 + W_2$ can be expressed in a unique way as

$$\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$$

with $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. In other words, show that if

$$\mathbf{w} = \mathbf{w}_1' + \mathbf{w}_2'$$

is another such expression, then we must have $\mathbf{w}_1 = \mathbf{w}_1'$ and $\mathbf{w}_2 = \mathbf{w}_2'$. [If $W_1 \cap W_2 = \{\mathbf{0}\}$, then $W_1 + W_2$ is written $W_1 \oplus W_2$ and is called the *direct sum* of W_1 and W_2.]
SOLUTION. Suppose $\mathbf{w} \in W_1 + W_2$ can be expressed in two different ways as in the problem statement. Then

$$\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{w}_1' + \mathbf{w}_2'.$$

and hence

$$\mathbf{w}_1 - \mathbf{w}_1' = \mathbf{w}_2' - \mathbf{w}_2. \tag{3.1}$$

小红书
小红书号:3189205503 qq

Let the vector in (3.1) be denoted by $\mathbf{v}$. Note that $\mathbf{v} = \mathbf{w}_1 - \mathbf{w}_1' \in W_1$ and $\mathbf{v} = \mathbf{w}_2' - \mathbf{w}_2 \in W_2$. So $\mathbf{v} \in W_1 \cap W_2$. Since $W_1 \cap W_2 = \{\mathbf{0}\}$, this means that $\mathbf{v} = \mathbf{0}$. Therefore, we have

$$\mathbf{w}_1 - \mathbf{w}_1' = \mathbf{0} \quad \text{and} \quad \mathbf{w}_2' - \mathbf{w}_2 = \mathbf{0}$$

and so

$$\mathbf{w}_1 = \mathbf{w}_1' \quad \text{and} \quad \mathbf{w}_2 = \mathbf{w}_2'.$$

4. (a) Consider the following elements of $\mathbb{R}^4$, the vector space of 4-tuples,

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ -1 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ -5 \\ -4 \\ 0 \end{bmatrix}.$$

Is $\mathbf{a}$ a linear combination of $\mathbf{b}$ and $\mathbf{c}$? If so, write $\mathbf{a}$ as a linear combination of $\mathbf{b}$ and $\mathbf{c}$.

(b) Consider the following elements of $\mathbb{R}^{2\times 2}$, the vector space of 2-by-2 matrices,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -5 \\ -4 & 0 \end{bmatrix}.$$

Is A a linear combination of B and C? If so, write A as a linear combination of B and C.

(c) Consider the following elements of $\mathbb{P}_3$, the vector space of polynomials of degree not more than 3,

$$a(x) = 1 + 2x + 3x^2 + x^3, \quad b(x) = 3 - x + 2x^2 + 2x^3, \quad c(x) = 1 - 5x - 4x^2.$$

Is $a(x)$ a linear combination of $b(x)$ and $c(x)$? If so, write $a(x)$ as a linear combination of $b(x)$ and $c(x)$.

SOLUTION. In all three cases, setting

$$\mathbf{a} = \lambda\mathbf{b} + \mu\mathbf{c}, \quad A = \lambda B + \mu C, \quad a(x) = \lambda b(x) + \mu c(x)$$

yield the same linear system of equations

$$\begin{cases} 3\lambda + \mu &= 1 \\ -\lambda - 5\mu &= 2 \\ 2\lambda - 4\mu &= 3 \\ 2\lambda &= 1 \end{cases}$$

which has a unique solution $\lambda = 1/2, \mu = -1/2$. The rest is straightforward.

5. Let $\mathbf{u} = (1, -3, 2)$ and $\mathbf{v} = (2, -1, 1) \in \mathbb{R}^3$.

(a) Is $(2, -5, 4) \in \mathbb{R}^3$ a linear combination of $\mathbf{u}$ and $\mathbf{v}$?

SOLUTION. Writing $(2, -5, 4) = x(1, -3, 2) + y(2, -1, 1)$ gives the following system of equations

$$\begin{cases} x + 2y &= 2 \\ -3x - y &= -5 \\ 2x + y &= 4 \end{cases} \quad \text{or} \quad \begin{cases} x + 2y &= 2 \\ 5y &= 1 \\ -3y &= 0 \end{cases} \quad \text{or} \quad \begin{cases} x + 2y &= 2 \\ 5y &= 1 \\ 0 &= \frac{3}{5} \end{cases}$$

which is inconsistent. Hence $(2, -5, 4)$ is not a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

(b) Write $(1, 7, -4) \in \mathbb{R}^3$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

SOLUTION. Writing $(1, 7, -4) = x(1, -3, 2) + y(2, -1, 1)$ gives the following system of equations

$$\begin{cases} x + 2y &= 1 \\ -3x - y &= 7 \\ 2x + y &= -4 \end{cases}$$

which has the solution $x = -3$ and $y = 2$. Hence $(1, 7, -4) = 3(1, -3, 2) + 2(2, -1, 1)$.

小红书
小红书号:3189205503 qq

(c) Find $a \in \mathbb{R}$ so that $(1, a, 5) \in \mathbb{R}^3$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.
SOLUTION. Writing $(1, a, 5) = x(1, -3, 2) + y(2, -1, 1)$ gives the following system of equations

$$\begin{cases} x + 2y & = 1 \\ -3x - y & = a \\ 2x + y & = 5 \end{cases}$$

solving the first and third equations gives us $x = 2, y = -1$; substituting these into the second equation gives

$$a = -8.$$

(d) Find conditions on $a, b, c \in \mathbb{R}$ so that $(a, b, c) \in \mathbb{R}^3$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.
SOLUTION. Writing $(a, b, c) = x(1, -3, 2) + y(2, -1, 1)$ gives the following system of equations

$$\begin{cases} x + 2y & = a \\ -3x - y & = b \\ 2x + y & = c \end{cases} \text{ or } \begin{cases} x + 2y & = a \\ 5y & = 3a + b \\ -3y & = -2a + c \end{cases} \text{ or } \begin{cases} x + 2y & = a \\ 5y & = 3a + b \\ 0 & = -a + 3b + 5c \end{cases}$$

which is consistent if and only if $a - 3b - 5c = 0$. Hence (a, b, c) is a linear combination of $\mathbf{u}$ and $\mathbf{v}$ iff $a - 3b - 5c = 0$.

小红书
小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 3

1. Let W, W_1, W_2 be subspaces of a vector space V. Recall from Problem Set 2 that

$$W \cap (W_1 + W_2) \neq (W \cap W_1) + (W \cap W_2)$$

in general. Show that the following statements are however true:

(a) $W \cap (W_1 + W_2) \supseteq (W \cap W_1) + (W \cap W_2)$.

SOLUTION. Since $W_1 \subseteq W_1 + W_2$, so $W \cap W_1 \subseteq W \cap (W_1 + W_2)$. Likewise, since $W_2 \subseteq W_1 + W_2$, so $W \cap W_2 \subseteq W \cap (W_1 + W_2)$. Hence

$$(W \cap W_1) \cup (W \cap W_2) \subseteq W \cap (W_1 + W_2).$$

Now since W and $W_1 + W_2$ are both subspaces of V, $W \cap (W_1 + W_2)$ is also a subspace of V. Now by Problem **3**(c) in Problem Set 2, the smallest subspace containing $(W \cap W_1) \cup (W \cap W_2)$ is $(W \cap W_1) + (W \cap W_2)$, which implies that

$$(W \cap W_1) + (W \cap W_2) \subseteq W \cap (W_1 + W_2). \tag{1.1}$$

(b) $W \cap (W_1 + (W \cap W_2)) = (W \cap W_1) + (W \cap W_2)$.

SOLUTION. Since

$$W \cap (W \cap W_2) = (W \cap W) \cap W_2 = W \cap W_2,$$

it follows from (a), with $W \cap W_2$ playing the role of W_2 in (1.1), that

$$(W \cap W_1) + (W \cap W_2) = (W \cap W_1) + (W \cap (W \cap W_2)) \subseteq W \cap (W_1 + (W \cap W_2)).$$

To show the reverse inclusion, let $\mathbf{w} \in W \cap (W_1 + (W \cap W_2))$. Then $\mathbf{w} \in W$ and $\mathbf{w} \in W_1 + (W \cap W_2)$. Hence $\mathbf{w} \in W$ and $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W \cap W_2$. Observe that $\mathbf{w}_2 \in W \cap W_2$ implies that $\mathbf{w}_2 \in W$; and noting that $\mathbf{w} \in W$, we get

$$\mathbf{w}_1 = \mathbf{w}_2 - \mathbf{w} \in W$$

since W is a subspace. Now $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_1 \in W$ together implies that $\mathbf{w}_1 \in W \cap W_1$. It then follows that $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2 \in (W \cap W_1) + (W \cap W_2)$. Since this is true for arbitrary $\mathbf{w} \in W \cap (W_1 + (W \cap W_2))$, we conclude that

$$W \cap (W_1 + (W \cap W_2)) \subseteq (W \cap W_1) + (W \cap W_2).$$

2. Let V be a vector space and $\mathbf{u}_1, \dots, \mathbf{u}_n \in V$.

(a) Suppose $\mathbf{w} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. Show that

$$\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}.$$

SOLUTION. Let $\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. Then

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n$$

for some $\lambda_1, \dots, \lambda_n \in \mathbb{F}$. But since $0\mathbf{w} = \mathbf{0}$, we could write

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n + 0\mathbf{w}$$

and so $\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}\}$. For the reverse inclusion, let $\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}\}$. Then

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n + \lambda_{n+1} \mathbf{w} \tag{2.2}$$

Date: October 18, 2007 (Version 1.0).

小红书
小红书号:3189205503 qq

for some $\lambda_1, \dots, \lambda_{n+1} \in \mathbb{F}$. Since $\mathbf{w} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$,

$$\mathbf{w} = \mu_1 \mathbf{u}_1 + \cdots + \mu_n \mathbf{u}_n \tag{2.3}$$

for some $\mu_1, \dots, \mu_n \in \mathbb{F}$. Substituting (2.3) into (2.2) yields

$$\begin{aligned} \mathbf{v} &= \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n + \lambda_{n+1}(\mu_1 \mathbf{u}_1 + \cdots + \mu_n \mathbf{u}_n) \\ &= (\lambda_1 + \lambda_{n+1}\mu_1)\mathbf{u}_1 + \cdots + (\lambda_n + \lambda_{n+1}\mu_1)\mathbf{u}_n \end{aligned}$$

and so $\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$.

(b) Generalize Problem **2**(a) to more than one vector. Suppose $\mathbf{w}_1, \dots, \mathbf{w}_m \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. Show that

$$\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_m\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}.$$

SOLUTION. We do this by induction. If $m = 1$, then it is true by (a). Suppose this is true for $m - 1$, ie.

$$\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_{m-1}\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}.$$

Then since $\mathbf{w}_m \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_{m-1}\}$, we may apply (a) again to get

$$\begin{aligned} \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_{m-1}, \mathbf{w}_m\} &= \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_{m-1}\} \\ &= \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}. \end{aligned}$$

Hence this is true for all $m \in \mathbb{N}$ by the priniciple of mathematical induction.

3. Let V be a vector space.

(a) Let W be a subspace of V. Show that $\operatorname{span}(W) = W$.

SOLUTION. Since W is a subspace of V, W is closed under linear combinations, ie. for any $\mathbf{u}_1, \dots, \mathbf{u}_n \in W$, any $\lambda_1, \dots, \lambda_n \in \mathbb{F}$, and any $n \in \mathbb{N}$,

$$\lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n \in W.$$

Since by its definition,

$$\operatorname{span}(W) = \{\lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n \in V \mid \mathbf{u}_1, \dots, \mathbf{u}_n \in W;\ \lambda_1, \dots, \lambda_n \in \mathbb{F};\ n \in \mathbb{N}\},$$

it follows that $\operatorname{span}(W) \subseteq W$. But clearly, $W \subseteq \operatorname{span}(W)$. Both inclusions then yield $\operatorname{span}(W) = W$.

(b) Let S be a subset of V. What is $\operatorname{span}(\operatorname{span}(S))$?

SOLUTION. Since $\operatorname{span}(S)$ is a subspace, it follows from (a) that

$$\operatorname{span}(\operatorname{span}(S)) = \operatorname{span}(S).$$

(c) Let S and T be subsets of V. Show that if $S \subseteq T$, then $\operatorname{span}(S) \subseteq \operatorname{span}(T)$.

SOLUTION. Let $\mathbf{v} \in \operatorname{span}(S)$. Then

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n$$

for some $\mathbf{u}_1, \dots, \mathbf{u}_n \in S$, $\lambda_1, \dots, \lambda_n \in \mathbb{F}$, and $n \in \mathbb{N}$. But $S \subseteq T$, so $\mathbf{u}_1, \dots, \mathbf{u}_n \in T$. Thus $\mathbf{v} \in \operatorname{span}(T)$. Accordingly, $\operatorname{span}(S) \subseteq \operatorname{span}(T)$.

(d) Let S be a subset of V. What is $\operatorname{span}(S \cup \{\mathbf{0}\})$?

SOLUTION. Since $S \subseteq S \cup \{\mathbf{0}\}$, it follows from (c) that $\operatorname{span}(S) \subseteq \operatorname{span}(S \cup \{\mathbf{0}\})$. Suppose $\mathbf{v} \in \operatorname{span}(S \cup \{\mathbf{0}\})$. Then

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n + \lambda_{n+1}\mathbf{0}$$

for some $\mathbf{u}_1, \dots, \mathbf{u}_n \in S$, $\lambda_1, \dots, \lambda_{n+1} \in \mathbb{F}$, and $n \in \mathbb{N}$. But since $\lambda_{n+1}\mathbf{0} = \mathbf{0}$, we have

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \cdots + \lambda_n \mathbf{u}_n$$

and so $\mathbf{v} \in \operatorname{span}(S)$. Thus $\operatorname{span}(S \cup \{\mathbf{0}\}) \subseteq \operatorname{span}(S)$. Both inclusions then yield

$$\operatorname{span}(S \cup \{\mathbf{0}\}) = \operatorname{span}(S).$$

小红书
小红书号:3189205503 qq

Alternatively, just apply Problem **2**(a) with $\mathbf{w} = \mathbf{0}$.

4. Let W_1, W_2 be subspaces of a vector space V. Recall from Problem Set 2 that $W = W_1 \oplus W_2$ if and only if (i) $W = W_1 + W_2$ and (ii) $W_1 \cap W_2 = \{\mathbf{0}\}$ are both satisfied.

(a) Let $V = \mathbb{R}^3$. Let $W_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}$ and $W_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x = 0\}$. Show that

$$\mathbb{R}^3 = W_1 \oplus W_2.$$

SOLUTION. Note that $(x, y, z) \in W_1 \cap W_2$ implies that $x = y = z$ and $x = 0$ and thus $x = 0, y = 0, z = 0$. Hence $W_1 \cap W_2 = \{(0, 0, 0)\}$. Now for any $(x, y, z) \in \mathbb{R}^3$, we may write

$$(x, y, z) = (x, x, x) + (0, y - x, z - x)$$

where $(x, x, x) \in W_1$ and $(0, y - x, z - x) \in W_2$. Hence $\mathbb{R}^3 = W_1 + W_2$. The two conditions $W_1 \cap W_2 = \{(0, 0, 0)\}$ and $\mathbb{R}^3 = W_1 + W_2$ imply that $\mathbb{R}^3 = W_1 \oplus W_2$.

(b) Let the vectors $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n \in V$ be linearly independent. Show that

$$\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m\} \oplus \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

SOLUTION. We know from Midterm 1, Problem **4**(b) that

$$\begin{aligned}\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m\} + \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} &= \operatorname{span}(\{\mathbf{u}_1, \dots, \mathbf{u}_m\} \cup \{\mathbf{v}_1, \dots, \mathbf{v}_n\}) \\ &= \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n\}.\end{aligned}$$

It remains to show that the sum is a direct sum (ie. we may write '$\oplus$' in place of '$+$'). In other words, we need the additional condition that

$$\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m\} \cap \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \{\mathbf{0}\}.$$

Suppose $\mathbf{w} \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m\} \cap \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Then we must have

$$\mathbf{w} = \lambda_1\mathbf{u}_1 + \cdots + \lambda_m\mathbf{u}_m \quad \text{and} \quad \mathbf{w} = \mu_1\mathbf{v}_1 + \cdots + \mu_n\mathbf{v}_n$$

for some $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ and $\mu_1, \dots, \mu_n \in \mathbb{F}$. So we get

$$\lambda_1\mathbf{u}_1 + \cdots + \lambda_m\mathbf{u}_m = \mu_1\mathbf{v}_1 + \cdots + \mu_n\mathbf{v}_n$$

and thus

$$\lambda_1\mathbf{u}_1 + \cdots + \lambda_m\mathbf{u}_m - \mu_1\mathbf{v}_1 - \cdots - \mu_n\mathbf{v}_n = \mathbf{0}.$$

Since $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly independent, this implies

$$\lambda_1 = \cdots = \lambda_m = \mu_1 = \cdots = \mu_n = 0$$

and thus $\mathbf{w} = 0\mathbf{u}_1 + \cdots + 0\mathbf{u}_m = \mathbf{0}$. Hence $\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m\} \cap \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \{\mathbf{0}\}$ as required.

5. Determine whether the following are linearly dependent.

(a) The vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^4$ where

$$\mathbf{a} = (1, 2, 3, 1), \quad \mathbf{b} = (3, -1, 2, 2), \quad \mathbf{c} = (1, -5, -4, 0).$$

SOLUTION. Similar to (c).

(b) The matrices $A, B, C \in \mathbb{R}^{2\times 2}$ where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -5 \\ -4 & 0 \end{bmatrix}.$$

SOLUTION. Similar to (c).

(c) The polynomials $a(x), b(x), c(x) \in \mathbb{P}_3$ where

$$a(x) = 1 + 2x + 3x^2 + x^3, \quad b(x) = 3 - x + 2x^2 + 2x^3, \quad c(x) = 1 - 5x - 4x^2.$$

SOLUTION. Suppose $\alpha, \beta, \gamma \in \mathbb{R}$ are such that

$$\alpha a(x) + \beta b(x) + \gamma c(x) = \mathbf{0},$$

小红书
小红书号:3189205503 qq

in other words,

$$\alpha(1+2x+3x^2+x^3)+\beta(3-x+2x^2+2x^3)+\gamma(1-5x-4x^2)=0+0x+0x^2+0x^3.$$

Then

$$(\alpha+3\beta+\gamma)+(2\alpha-\beta-5\gamma)x+(3\alpha+2\beta-4\gamma)x^2+(\alpha+2\beta)x^3=0+0x+0x^2+0x^3.$$

Equating corresponding coefficients gives the system

$$\begin{aligned}\alpha+3\beta+\gamma&=0,\\ 2\alpha-\beta-5\gamma&=0,\\ 3\alpha+2\beta-4\gamma&=0,\\ \alpha+2\beta&=0.\end{aligned}$$

Applying Gaussian Elimination, we may reduce this to

$$\begin{aligned}\alpha+3\beta+\gamma&=0,\\ \beta+\gamma&=0.\end{aligned}$$

So this solution to system involves a free parameter and hence there exists nonzero solutions. For example, $\alpha=2,\beta=-1,\gamma=1$, which gives

$$2a(x)-b(x)+c(x)=\mathbf{0}.$$

So the polynomials $a(x),b(x),c(x)$ are linearly dependent.

(d) The vectors $\mathbf{p},\mathbf{q}\in\mathbb{R}^6$ where

$$\mathbf{p}=(1,2,-3,4,-5,6),\quad \mathbf{q}=(6,-5,4,-3,2,1).$$

SOLUTION. Similar to (e).

(e) The matrices $P,Q\in\mathbb{R}^{2\times3}$ where

$$P=\begin{bmatrix}1&2&-3\\6&-5&4\end{bmatrix},\quad Q=\begin{bmatrix}6&-5&4\\1&2&-3\end{bmatrix}.$$

SOLUTION. Suppose $\lambda,\mu\in\mathbb{R}$ are such that

$$\lambda P+\mu Q=\mathbf{0},$$

in other words,

$$\lambda\begin{bmatrix}1&2&-3\\6&-5&4\end{bmatrix}+\mu\begin{bmatrix}6&-5&4\\1&2&-3\end{bmatrix}=\begin{bmatrix}0&0&0\\0&0&0\end{bmatrix}.$$

Then

$$\begin{bmatrix}\lambda+6\mu&2\lambda-5\mu&-3\lambda+4\mu\\6\lambda+\mu&-5\lambda+2\mu&4\lambda-3\mu\end{bmatrix}=\begin{bmatrix}0&0&0\\0&0&0\end{bmatrix}.$$

Equating corresponding entries gives the system

$$\begin{aligned}\lambda+6\mu&=0,\\ 2\lambda-5\mu&=0,\\ -3\lambda+4\mu&=0,\\ 6\lambda+\mu&=0,\\ -5\lambda+2\mu&=0,\\ 4\lambda-3\mu&=0.\end{aligned}$$

This has the unique solution $\lambda=0$, $\mu=0$. Hence P and Q are linearly independent.

(f) The polynomials $p(x),q(x)\in\mathbb{P}_5$ where

$$a(x)=1+2x-3x^2+4x^3-5x^4+6x^5,\quad b(x)=6-5x+4x^2-3x^3+2x^4+x^5.$$

SOLUTION. Similar to (e).

小红书
小红书号:3189205503 qq

6. Let V be a vector space and $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.

(a) If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent, show that $\mathbf{u}+\mathbf{v}$, $\mathbf{u}-\mathbf{v}$, and $\mathbf{u}-2\mathbf{v}+\mathbf{w}$ are also linearly independent.

SOLUTION. Suppose

$$\mu_1(\mathbf{u}+\mathbf{v}) + \mu_2(\mathbf{u}-\mathbf{v}) + \mu_3(\mathbf{u}-2\mathbf{v}+\mathbf{w}) = \mathbf{0}$$

for some $a, b, c \in \mathbb{F}$. Therefore,

$$(\mu_1+\mu_2+\mu_3)\mathbf{u} + (\mu_1-\mu_2-2\mu_3)\mathbf{v} + \mu_3\mathbf{w} = \mathbf{0}.$$

Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent, we must have

$$\begin{aligned} \mu_1+\mu_2+\mu_3 &= 0, \\ \mu_1-\mu_2-2\mu_3 &= 0, \\ \mu_3 &= 0. \end{aligned}$$

The only solution to the above system is $\mu_1 = 0, \mu_2 = 0, \mu_3 = 0$. Hence $\mathbf{u}+\mathbf{v}$, $\mathbf{u}-\mathbf{v}$, and $\mathbf{u}-2\mathbf{v}+\mathbf{w}$ are linearly independent.

(b) If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent and they span V, show that $\mathbf{u}+\mathbf{v}$, $\mathbf{u}-\mathbf{v}$, and $\mathbf{u}-2\mathbf{v}+\mathbf{w}$ also span V.

SOLUTION. Let $\mathbf{x} \in V$. Then since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ span V, there exist $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{F}$ such that

$$\mathbf{x} = \lambda_1\mathbf{u} + \lambda_2\mathbf{v} + \lambda_3\mathbf{w}.$$

We claim that there also exist $\mu_1, \mu_2, \mu_3 \in \mathbb{F}$ such that

$$\mathbf{x} = \mu_1(\mathbf{u}+\mathbf{v}) + \mu_2(\mathbf{u}-\mathbf{v}) + \mu_3(\mathbf{u}-2\mathbf{v}+\mathbf{w}).$$

In other words, the equation

$$\mu_1(\mathbf{u}+\mathbf{v}) + \mu_2(\mathbf{u}-\mathbf{v}) + \mu_3(\mathbf{u}-2\mathbf{v}+\mathbf{w}) = \lambda_1\mathbf{u} + \lambda_2\mathbf{v} + \lambda_3\mathbf{w} \tag{6.4}$$

has a solution for μ_1, μ_2, μ_3 in terms of $\lambda_1, \lambda_2, \lambda_3$. Now, (6.4) may be rewritten as

$$(\mu_1+\mu_2+\mu_3)\mathbf{u} + (\mu_1-\mu_2-2\mu_3)\mathbf{v} + \mu_3\mathbf{w} = \lambda_1\mathbf{u} + \lambda_2\mathbf{v} + \lambda_3\mathbf{w},$$

$$(\mu_1+\mu_2+\mu_3-\lambda_1)\mathbf{u} + (\mu_1-\mu_2-2\mu_3-\lambda_2)\mathbf{v} + (\mu_3-\lambda_3)\mathbf{w} = \mathbf{0}.$$

Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent, we must have

$$\begin{aligned} \mu_1+\mu_2+\mu_3 &= \lambda_1, \\ \mu_1-\mu_2-2\mu_3 &= \lambda_2, \\ \mu_3 &= \lambda_3, \end{aligned}$$

which can be solved to give

$$\mu_1 = (\lambda_1+\lambda_2+\lambda_3)/2, \quad \mu_2 = (\lambda_1-\lambda_2-3\lambda_3)/2, \quad \mu_3 = \lambda_3.$$

Hence $\mathbf{x} = \lambda_1\mathbf{u} + \lambda_2\mathbf{v} + \lambda_3\mathbf{w} \in V$ can be rewritten as

$$\mathbf{x} = \frac{1}{2}(\lambda_1+\lambda_2+\lambda_3)(\mathbf{u}+\mathbf{v}) + \frac{1}{2}(\lambda_1-\lambda_2-3\lambda_3)(\mathbf{u}-\mathbf{v}) + \lambda_3(\mathbf{u}-2\mathbf{v}+\mathbf{w}).$$

Since $\mathbf{x} \in V$ is arbitrary, it follows that $\mathbf{u}+\mathbf{v}$, $\mathbf{u}-\mathbf{v}$, and $\mathbf{u}-2\mathbf{v}+\mathbf{w}$ span V.

小红书
小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 4 SOLUTIONS

For a matrix $A = [a_{ij}]_{i,j=1}^{m,n} \in \mathbb{F}^{m\times n}$, the *transpose* of A is the matrix $A^\top = [a_{ji}]_{j,i=1}^{n,m} \in \mathbb{F}^{n\times m}$. A square matrix $A \in \mathbb{R}^{n\times n}$ is called *symmetric* if $a_{ji} = a_{ij}$ for all $i, j \in \{1, \dots, n\}$ and is called *skew-symmetric* or *anti-symmetric* if $a_{ji} = -a_{ij}$ for all $i, j \in \{1, \dots, n\}$.
A basis will be denoted $\mathscr{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ when the ordering of the basis vectors is not important and $\mathscr{B} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n]$ when it is.

1. Find a basis for the following vector spaces
(a) $\mathbb{C}^{n\times n}$ as a vector space over $\mathbb{C}$.
(b) $\mathbb{C}^{n\times n}$ as a vector space over $\mathbb{R}$.
(c) $\mathsf{S}^2(\mathbb{R}^n) = \{A \in \mathbb{R}^{n\times n} \mid A^\top = A\}$, ie. the real symmetric matrices, as a vector space over $\mathbb{R}$.
(d) $\wedge^2(\mathbb{R}^n) = \{A \in \mathbb{R}^{n\times n} \mid A^\top = -A\}$, ie. the real skew-symmetric matrices, as a vector space over $\mathbb{R}$.

Show that

$$\mathbb{R}^{n\times n} = \mathsf{S}^2(\mathbb{R}^n) \oplus \wedge^2(\mathbb{R}^n).$$

SOLUTION. Let E_{st} denote the $n \times n$ matrix with the (s,t)-entry 1 and 0 everywhere else, $s, t = 1, 2, \dots, n$. A basis for $\mathbb{C}^{n\times n}$ as a vector space over $\mathbb{C}$ is given by $\{E_{st} \mid s, t = 1, \dots, n\}$. A basis for $\mathbb{C}^{n\times n}$ as a vector space over $\mathbb{R}$ is given by $\{E_{st} \mid s, t = 1, \dots, n\} \cup \{iE_{st} \mid s, t = 1, \dots, n\}$. A basis for $\mathsf{S}^2(\mathbb{R}^n)$ as a vector space over $\mathbb{R}$ is given by $\{E_{st} + E_{ts} \mid s \le t,\ s, t = 1, \dots, n\}$. A basis for $\wedge^2(\mathbb{R}^n)$ as a vector space over $\mathbb{R}$ is given by $\{E_{st} - E_{ts} \mid s < t,\ s, t = 1, \dots, n\}$. Since every matrix $A \in \mathbb{R}^{n\times n}$ can be written

$$A = \frac{1}{2}(A + A^\top) + \frac{1}{2}(A - A^\top)$$

and we may check that

$$\frac{1}{2}(A + A^\top) \in \mathsf{S}^2(\mathbb{R}^n), \quad \frac{1}{2}(A - A^\top) \in \wedge^2(\mathbb{R}^n),$$

this shows that $\mathbb{R}^{n\times n} = (\mathsf{S}^2(\mathbb{R}^n)) + (\wedge^2(\mathbb{R}^n))$. Now if $B \in \mathsf{S}^2(\mathbb{R}^n)$ and $B \in \wedge^2(\mathbb{R}^n)$, then

$$B = B^\top = -B$$

which can only happen if $B = \mathbf{0}$. So $(\mathsf{S}^2(\mathbb{R}^n)) \cap (\wedge^2(\mathbb{R}^n)) = \{\mathbf{0}\}$. Hence

$$\mathbb{R}^{n\times n} = \mathsf{S}^2(\mathbb{R}^n) \oplus \wedge^2(\mathbb{R}^n).$$

2. Let V be a vector space and let $\mathscr{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for V. Suppose $n \ge 2$. Let

$$\mathscr{B}' = \{\mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_2 + \mathbf{u}_3, \dots, \mathbf{u}_{n-1} + \mathbf{u}_n, \mathbf{u}_n + \mathbf{u}_1\}.$$

Is $\mathscr{B}'$ also a basis for V? How about the converse? ie. if $\mathscr{B}'$ is a basis for V, is $\mathscr{B}$ necessarily a basis for V? [Hint: the answer depends on whether n is odd or even.]
SOLUTION. We will check for linear independence. Let $\lambda_1, \dots, \lambda_n$ be such that

$$\lambda_1(\mathbf{u}_1 + \mathbf{u}_2) + \lambda_2(\mathbf{u}_2 + \mathbf{u}_3) + \cdots + \lambda_{n-1}(\mathbf{u}_{n-1} + \mathbf{u}_n) + \lambda_n(\mathbf{u}_n + \mathbf{u}_1) = \mathbf{0}.$$

Therefore

$$(\lambda_n + \lambda_1)\mathbf{u}_1 + (\lambda_1 + \lambda_2)\mathbf{u}_2 + \cdots + (\lambda_{n-2} + \lambda_{n-1})\mathbf{u}_{n-1} + (\lambda_{n-1} + \lambda_n)\mathbf{u}_n = \mathbf{0}.$$

Date: October 18, 2007 (Version 1.1).

小红书
小红书号:3189205503 qq

Since $\mathscr{B}$ is linearly independent, we must have

$$\lambda_n + \lambda_1 = 0, \lambda_1 + \lambda_2 = 0, \ldots, \lambda_{n-1} + \lambda_n = 0.$$

This is a homogeneous system of linear equation. Applying Gaussian Elimination to the corresponding coefficient matrix, we get

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2 : R_2 - R_1} \left[\begin{array}{ccccccc|c} 1 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 : R_3 - R_2}$$

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_4 : R_4 - R_3} \quad \cdots \quad \xrightarrow{R_{n-1} : R_{n-1} - R_{n-2}}$$

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 & (-1)^n & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_n : R_n - R_{n-1}} \left[\begin{array}{ccccccc|c} 1 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 & (-1)^n & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1-(-1)^n & 0 \end{array}\right].$$

Now, whether the system has a unique solution or not depends on whether the last entry $1-(-1)^n$ is zero or not. If n is even, then $1-(-1)^n = 0$ and the system does not have a unique solution — in particular, it must have a solution where not all $\lambda_1, \ldots, \lambda_n$ are zero. Hence $\mathscr{B}'$ is linearly dependent and thus not a basis. If n is odd, then $1-(-1)^n = 2 \neq 0$ and the system has a unique solution, ie. $\lambda_1 = 0, \lambda_2 = 0, \ldots, \lambda_n = 0$ and so $\mathscr{B}'$ is linearly independent. Since $\mathscr{B}'$ has n elements and $\dim(V) = |\mathscr{B}| = n$, it is a basis of V. Alternatively, we could have observed that if n is odd then the row-reduced echelon form of the coefficient matrix is

$$\left[\begin{array}{ccccccc} 1 & 0 & 0 & \cdots & 0 & 0 & 1 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -1 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{array}\right]$$

which means that

$$\lambda_1(\mathbf{u}_1 + \mathbf{u}_2) + \lambda_2(\mathbf{u}_2 + \mathbf{u}_3) + \cdots + \lambda_{n-1}(\mathbf{u}_{n-1} + \mathbf{u}_n) + \lambda_n(\mathbf{u}_n + \mathbf{u}_1) = \mathbf{v} \tag{2.1}$$

will always have a solution for any $\mathbf{v} \in V$, ie. every $\mathbf{v} \in V$ may be written as a linear combination of $\mathscr{B}'$. Hence $\mathscr{B}'$ is a basis for V when n is odd. The converse is true. Suppose $\mathscr{B}'$ is a basis of V. Since each vector in $\mathscr{B}'$ is a linear combination of vectors in $\mathscr{B}$, $\mathscr{B}'$ spans V implies that $\mathscr{B}$ spans V; and since $|\mathscr{B}| = |\mathscr{B}'| = \dim(V)$, $\mathscr{B}$ must also be a basis.

3. For the following real vector spaces V, fnd the coordinate representation of the element $\mathbf{v}$ with respect to the ordered basis $\mathscr{B}$.

(a) $V = \mathbb{P}_2 = \{a + bx + cx^2 \mid a, b, c \in \mathbb{R}\}$. Find $[\mathbf{v}]_{\mathscr{B}}$ where

$$\mathbf{v} = 6 - 5x + 2x^2, \quad \mathscr{B} = [1, -1 + x, 1 - 2x + x^2].$$

小红书
小红书号:3189205503 qq

SOLUTION. Set $\mathbf{v}$ as a linear combination of elements in the basis:

$$\begin{aligned} 6 - 5x + 2x^2 &= \alpha + \beta(-1 + x) + \gamma(1 - 2x + x^2) \\ &= (\alpha - \beta + \gamma) + (\beta - 2\gamma)x + \gamma x^2. \end{aligned}$$

We equate coefficients of corresponding powers of x and solve the resulting linear system of equation

$$\begin{aligned} \alpha - \beta + \gamma &= 6, \\ \beta - 2\gamma &= -5, \\ \gamma &= 2, \end{aligned}$$

to get $\alpha = 3, \beta = -1, \gamma = 2$. Hence

$$[6 - 5x + 2x^2]_{\mathscr{B}} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

(b) $V = \mathsf{S}^2(\mathbb{R}^2)$. Find $[\mathbf{v}]_{\mathscr{B}}$ where

$$\mathbf{v} = \begin{pmatrix} 4 & -11 \\ -11 & -7 \end{pmatrix}, \quad \mathscr{B} = \left[\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 4 & -1 \\ -1 & -5 \end{pmatrix} \right].$$

SOLUTION. Set $\mathbf{v}$ as a linear combination of elements in the basis:

$$\begin{aligned} \begin{pmatrix} 4 & -11 \\ -11 & -7 \end{pmatrix} &= x \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} + y \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} + z \begin{pmatrix} 4 & -1 \\ -1 & -5 \end{pmatrix} \\ &= \begin{pmatrix} x + 2y + 4z & -2x + y - z \\ -2x + y - z & x + 3y - 5z \end{pmatrix}. \end{aligned}$$

We equate the correponding matrix entries and solve the resulting linear system of equation

$$\begin{aligned} x + 2y + 4z &= 4, \\ -2x + y - z &= -11, \\ x + 3y - 5z &= -7, \end{aligned}$$

to get $x = 4, y = -2, z = 1$. Hence

$$\left[\begin{pmatrix} 4 & -11 \\ -11 & -7 \end{pmatrix} \right]_{\mathscr{B}} = \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix}.$$

(c) $V = \mathbb{P}_3 = \{a + bx + cx^2 + dx^3 \mid a, b, c, d \in \mathbb{R}\}$. Find $[\mathbf{v}]_{\mathscr{B}}$ where

$$\mathbf{v} = a + bx + cx^2 + dx^3, \quad \mathscr{B} = [1, (1 - x), (1 - x)^2, (1 - x)^3].$$

SOLUTION. Set $\mathbf{v}$ as a linear combination of elements in the basis:

$$\begin{aligned} a + bx + cx^2 + dx^3 &= \alpha + \beta(1 - x) + \gamma(1 - x)^2 + \delta(1 - x)^3 \\ &= (\alpha + \beta + \gamma + \delta) + (-\beta - 2\gamma - 3\delta)x + (\gamma + 3\delta)x^2 - \delta x^3. \end{aligned}$$

We equate coefficients of corresponding powers of x and solve the resulting linear system of equation

$$\begin{aligned} \alpha + \beta + \gamma + \delta &= a, \\ -\beta - 2\gamma - 3\delta &= b, \\ \gamma + 3\delta &= c, \\ -\delta &= d, \end{aligned}$$

小红书
小红书号:3189205503 qq

to get

$$\begin{aligned}\alpha &= a+b+c+d,\\ \beta &= -b-2c-3d,\\ \gamma &= c+3d,\\ \delta &= -d.\end{aligned}$$

Hence

$$[a+bx+cx^2+dx^3]_{\mathscr{B}} = \begin{bmatrix} a+b+c+d \\ -b-2c-3d \\ c+3d \\ -d \end{bmatrix}.$$

4. Let W_1, W_2 be subspaces of a vector space V.

(a) Show that

$$\dim(W_1+W_2)+\dim(W_1\cap W_2) = \dim(W_1)+\dim(W_2). \tag{4.2}$$

SOLUTION. Let $\dim(W_1\cap W_2)=k$ and let $\{\mathbf{u}_1,\dots,\mathbf{u}_k\}$ be a basis for $W_1\cap W_2$. Note that $\{\mathbf{u}_1,\dots,\mathbf{u}_k\}$ is a linearly independent subset of W_1 and by the Adding-On Algorithm, we may add vectors $\mathbf{v}_1,\dots,\mathbf{v}_m$ to it so that

$$\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{v}_1,\dots,\mathbf{v}_m\}$$

is a basis for W_1. Applying the same argument now to W_2, we may also add vectors $\mathbf{w}_1,\dots,\mathbf{w}_n$ so that

$$\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{w}_1,\dots,\mathbf{w}_n\}$$

is a basis for W_2. Hence

$$\dim(W_1)=k+m, \quad \dim(W_2)=k+n.$$

From a result that proved in the Mid-Term I, we know that

$$\begin{aligned}W_1+W_2 &= \operatorname{span}(\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{v}_1,\dots,\mathbf{v}_m\}\cup\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{w}_1,\dots,\mathbf{w}_n\})\\ &= \operatorname{span}\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{v}_1,\dots,\mathbf{v}_m,\mathbf{w}_1,\dots,\mathbf{w}_n\}.\end{aligned}$$

We will show that $\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{v}_1,\dots,\mathbf{v}_m,\mathbf{w}_1,\dots,\mathbf{w}_n\}$ is linearly independent. Suppose

$$\alpha_1\mathbf{u}_1+\cdots+\alpha_k\mathbf{u}_k+\beta_1\mathbf{v}_1+\cdots+\beta_m\mathbf{v}_m+\gamma_1\mathbf{w}_1+\cdots+\gamma_n\mathbf{w}_n=\mathbf{0}. \tag{4.3}$$

We let

$$\mathbf{v}=\alpha_1\mathbf{u}_1+\cdots+\alpha_k\mathbf{u}_k+\beta_1\mathbf{v}_1+\cdots+\beta_m\mathbf{v}_m\in W_1. \tag{4.4}$$

Then (4.3) implies that

$$\mathbf{v}=-\gamma_1\mathbf{w}_1-\cdots-\gamma_n\mathbf{w}_n\in W_2. \tag{4.5}$$

So $\mathbf{v}\in W_1\cap W_2$ and since this has $\{\mathbf{u}_1,\dots,\mathbf{u}_k\}$ as a basis, there exist $\lambda_1,\dots,\lambda_k$ such that

$$\mathbf{v}=\lambda_1\mathbf{u}_1+\cdots+\lambda_k\mathbf{u}_k. \tag{4.6}$$

Now (4.5) and (4.6) together implies that

$$\lambda_1\mathbf{u}_1+\cdots+\lambda_k\mathbf{u}_k+\gamma_1\mathbf{w}_1+\cdots+\gamma_n\mathbf{w}_n=\mathbf{0};$$

and since $\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{w}_1,\dots,\mathbf{w}_n\}$ is linearly independent, we get

$$\lambda_1=\cdots=\lambda_k=\gamma_1=\cdots=\gamma_n=0.$$

Substituting $\gamma_1=\cdots=\gamma_n=0$ into (4.3) gives

$$\alpha_1\mathbf{u}_1+\cdots+\alpha_k\mathbf{u}_k+\beta_1\mathbf{v}_1+\cdots+\beta_m\mathbf{v}_m=\mathbf{0};$$

and since $\{\mathbf{u}_1,\dots,\mathbf{u}_k,\mathbf{v}_1,\dots,\mathbf{v}_n\}$ is linearly independent, we get

$$\alpha_1=\cdots=\alpha_k=\beta_1=\cdots=\beta_m=0.$$

小红书
小红书号:3189205503 qq

Hence this shows that $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n\}$ is linearly independent and is thus a basis for $W_1 + W_2$. Therefore $\dim(W_1 + W_2) = k + m + n$. Finally, we have

$$\begin{aligned}\dim(W_1 + W_2) + \dim(W_1 \cap W_2) &= (k + m + n) + k \\ &= (k + m) + (k + n) \\ &= \dim(W_1) + \dim(W_2).\end{aligned}$$

(b) Let $\mathscr{B} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a basis for V. Suppose W is a k-dimensional subspace of V. Show that for any subset $\{\mathbf{u}_{i_1}, \mathbf{u}_{i_2}, \ldots, \mathbf{u}_{i_m}\} \subseteq \mathscr{B}$ with $m > n - k$, there exists a nonzero vector $\mathbf{w} \in W$ that is a linear combination of $\mathbf{u}_{i_1}, \mathbf{u}_{i_2}, \ldots, \mathbf{u}_{i_m}$.
SOLUTION. Let $W_1 = W$ and $W_2 = \operatorname{span}\{\mathbf{u}_{i_1}, \mathbf{u}_{i_2}, \ldots, \mathbf{u}_{i_m}\}$. Since any subset of a linearly independent set (in this case $\mathscr{B}$) must be linearly independent, $\{\mathbf{u}_{i_1}, \mathbf{u}_{i_2}, \ldots, \mathbf{u}_{i_m}\}$ is linearly independent and so is a basis of W_2. Hence $\dim(W_2) = m$. Now $W_1 + W_2$ is a subspace of V and so

$$\dim(W_1 + W_2) \leq \dim(V) = n.$$

From (4.2), we have

$$\dim(W_1 \cap W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 + W_2) \geq k + m - n > 0$$

since $m > n - k$. In other words, $W_1 \cap W_2$ cannot be the zero vector space (otherwise, we will have $\dim(W_1 \cap W_2) = 0$). Hence $W_1 \cap W_2$ must contain a nonzero vector $\mathbf{w}$. By our definitions of W_1 and W_2, it follows that $\mathbf{w} \in W$ and is a linear combination of $\mathbf{u}_{i_1}, \mathbf{u}_{i_2}, \ldots, \mathbf{u}_{i_m}$.

(c) Find an expression for $\dim(W_1 \oplus W_2)$ similar to (4.2) (assuming that it makes sense to write $W_1 \oplus W_2$).
SOLUTION. Recall that we write $W_1 \oplus W_2$ for $W_1 + W_2$ when $W_1 \cap W_2 = \{\mathbf{0}\}$. So by (4.2),

$$\dim(W_1 \oplus W_2) + \dim(\{\mathbf{0}\}) = \dim(W_1) + \dim(W_2),$$

and since $\dim(\{\mathbf{0}\}) = 0$,

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2).$$

(d) Let W be a subspace of V having the property that there exists a *unique* subspace W' such that $V = W \oplus W'$. Show that $W = V$.
SOLUTION. Let $\dim(W) = r$ and $\dim(V) = n$. Suppose $W \neq V$. Then $r < n$. Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ be a basis for W. By the Adding-On Algorithm, we may add vectors $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_n$ such that

$$\{\mathbf{u}_1, \ldots, \mathbf{u}_r, \mathbf{u}_{r+1}, \ldots, \mathbf{u}_n\}$$

is a basis for V. Let

$$W' = \operatorname{span}\{\mathbf{u}_{r+1}, \mathbf{u}_{r+2}, \ldots, \mathbf{u}_n\}$$

and

$$W'' = \operatorname{span}\{\mathbf{u}_1 + \mathbf{u}_{r+1}, \mathbf{u}_{r+2}, \ldots, \mathbf{u}_n\}.$$

Note that $W' \neq W''$ — otherwise, we will have $\mathbf{u}_1 \in W'$, a contradiction. Furthermore, it follows from Homework **3**, Problem **4**(b), that

$$V = W \oplus W' \quad \text{and} \quad V = W \oplus W'',$$

which means that the decomposition is not unique. So for the decomposition to be unique, we must have $W = V$.

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小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 5

1. Let W_1 and W_2 be the following subspaces of $\mathbb{R}^4$.

$$W_1 = \operatorname{span}\left\{\begin{bmatrix}1\\2\\-1\\-2\end{bmatrix}, \begin{bmatrix}3\\1\\1\\1\end{bmatrix}, \begin{bmatrix}-1\\0\\1\\-1\end{bmatrix}\right\}, \quad W_2 = \operatorname{span}\left\{\begin{bmatrix}2\\5\\-6\\-5\end{bmatrix}, \begin{bmatrix}-1\\2\\-7\\3\end{bmatrix}\right\}.$$

(a) Find a basis and state the dimension of $W_1 \cap W_2$.

SOLUTION. Let the three vectors spanning W_1 be $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ and the two vectors spanning W_2 be $\mathbf{v}_1, \mathbf{v}_2$ respectively. If $\mathbf{w} \in W_1 \cap W_2$, then there exist $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ and $\beta_1, \beta_2 \in \mathbb{R}$ such that

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \alpha_3\mathbf{u}_3 = \mathbf{w} = \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2.$$

In other words, we want to solve

$$\alpha_1\begin{bmatrix}1\\2\\-1\\-2\end{bmatrix} + \alpha_2\begin{bmatrix}3\\1\\1\\1\end{bmatrix} + \alpha_3\begin{bmatrix}-1\\0\\1\\-1\end{bmatrix} = \beta_1\begin{bmatrix}2\\5\\-6\\-5\end{bmatrix} + \beta_2\begin{bmatrix}-1\\2\\-7\\3\end{bmatrix}.$$

Forming the augmented system to solve for $\alpha_1, \alpha_2, \alpha_3$ in terms of β_1, β_2 (alternatively, we could also solve for β_1 and β_2 in terms of $\alpha_1, \alpha_2, \alpha_3$), we get upon Gauss-Jordan elimination,

$$\left[\begin{array}{ccc|c} 1 & 3 & -1 & 2\beta_1 - \beta_2 \\ 2 & 1 & 0 & 5\beta_1 + 2\beta_2 \\ -1 & 1 & 1 & -6\beta_1 - 7\beta_2 \\ -2 & 1 & -1 & -5\beta_1 + 3\beta_2 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3\beta_1 + 2\beta_2 \\ 0 & 1 & 0 & -\beta_1 - 2\beta_2 \\ 0 & 0 & 1 & -2\beta_1 - 9\beta_2 \\ 0 & 0 & 0 & -6\beta_2 \end{array}\right].$$

For this system to be consistent, we must have $\beta_2 = 0$. Hence

$$\mathbf{w} = \beta_1[2, 5, -6, -5]^\top$$

for some $\beta_2 \in \mathbb{R}$. In other words,

$$W_1 \cap W_2 = \operatorname{span}\{[2, 5, -6, -5]^\top\}$$

and so $\dim(W_1 \cap W_2) = 1$. As a sanity check, observe that $\beta_2 = 0$ gives

$$\alpha_1 = 3\beta_1, \quad \alpha_2 = -\beta_1, \quad \alpha_3 = -2\beta_1$$

and indeed $\mathbf{w}$ can also be expressed as

$$\mathbf{w} = 3\beta_1\begin{bmatrix}1\\2\\-1\\-2\end{bmatrix} - \beta_1\begin{bmatrix}3\\1\\1\\1\end{bmatrix} - 2\beta_1\begin{bmatrix}-1\\0\\1\\-1\end{bmatrix} \in W_1.$$

(b) Find a basis and state the dimension of $W_1 + W_2$.

SOLUTION. Observe that

$$\operatorname{span}(S_1) + \operatorname{span}(S_2) = \operatorname{span}(S_1 \cup S_2).$$

So

$$W_1 + W_2 = \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{v}_1, \mathbf{v}_2\}.$$

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小红书
小红书号:3189205503 qq

We will apply the Throwing-Out Algorithm to get a basis. First we try to solve

$$\mathbf{v}_2 = \alpha\mathbf{u}_1 + \beta\mathbf{u}_2 + \gamma\mathbf{u}_3 + \delta\mathbf{v}_1. \tag{1.1}$$

Applying Gauss-Jordan Elimination, we get

$$\left[\begin{array}{cccc|c} 1 & 3 & -1 & 2 & -1 \\ 2 & 1 & 0 & 5 & 2 \\ -1 & 1 & 1 & -6 & -7 \\ -2 & 1 & -1 & -5 & 3 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & -3 & 2 \\ 0 & 1 & 0 & 1 & -2 \\ 0 & 0 & 1 & 2 & -9 \\ 0 & 0 & 0 & 0 & -6 \end{array}\right]$$

which shows that (1.1) has no solution — so we cannot discard $\mathbf{v}_2$. Next we try to solve

$$\mathbf{v}_1 = \alpha\mathbf{u}_1 + \beta\mathbf{u}_2 + \gamma\mathbf{u}_3 + \delta\mathbf{v}_2. \tag{1.2}$$

Applying Gauss-Jordan Elimination, we get

$$\left[\begin{array}{cccc|c} 1 & 3 & -1 & 2 & 2 \\ 2 & 1 & 2 & 5 & 5 \\ -1 & 1 & -7 & -6 & -6 \\ -2 & 1 & 3 & -5 & -5 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

which shows that (1.2) has a solution (given by $\alpha = 3, \beta = -1, \gamma = -2, \delta = 0$). Hence we may discard $\mathbf{v}_1$. Next we try to solve

$$\mathbf{u}_3 = \alpha\mathbf{u}_1 + \beta\mathbf{u}_2 + \gamma\mathbf{v}_2. \tag{1.3}$$

Applying Gauss-Jordan Elimination, we get

$$\left[\begin{array}{ccc|c} 1 & 3 & -1 & -1 \\ 2 & 1 & 2 & 0 \\ -1 & 1 & -7 & 1 \\ -2 & 1 & 3 & -1 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{array}\right]$$

which shows that (1.3) has no solution — so we cannot discard $\mathbf{u}_3$. Next we try to solve

$$\mathbf{u}_2 = \alpha\mathbf{u}_1 + \beta\mathbf{u}_3 + \gamma\mathbf{v}_2. \tag{1.4}$$

Applying Gauss-Jordan Elimination, we get

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 3 \\ 2 & 0 & 2 & 1 \\ -1 & 1 & -7 & 1 \\ -2 & -1 & 3 & 1 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{array}\right]$$

which shows that (1.4) has no solution — so we cannot discard $\mathbf{u}_2$. Lastly we try to solve

$$\mathbf{u}_1 = \alpha\mathbf{u}_2 + \beta\mathbf{u}_3 + \gamma\mathbf{v}_2. \tag{1.5}$$

Applying Gauss-Jordan Elimination, we get

$$\left[\begin{array}{ccc|c} 3 & -1 & -1 & 1 \\ 1 & 0 & 2 & 2 \\ 1 & 1 & -7 & -1 \\ 1 & -1 & 3 & -2 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -\frac{11}{2} \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & 11 \end{array}\right]$$

which shows that (1.5) has no solution — so we cannot discard $\mathbf{u}_1$. Hence by the Throwing-Out Algorithm the set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{v}_2\}$ is a basis for $W_1 + W_2$ and so $\dim(W_1 + W_2) = 4$. [Note: We will soon learn a more efficient method to determine a basis from a spanning set of vectors.]

2. Let $\mathbb{F}$ be a field. Let W_1 and W_2 be the following subspaces of $\mathbb{F}^n$.

$$W_1 = \{(x_1, x_2, \ldots, x_n) \in \mathbb{F}^n \mid x_1 + x_2 + \cdots + x_n = 0\},$$
$$W_2 = \{(x_1, x_2, \ldots, x_n) \in \mathbb{F}^n \mid x_1 = x_2 = \cdots = x_n\}.$$

小红书
小红书号:3189205503 qq

(a) Find a basis and state the dimension of W_1.
SOLUTION. Let $(x_1, x_2, \ldots, x_n) \in W_1$. Then

$$(x_1, x_2, \ldots, x_n) = (x_1, x_2, \ldots, -x_1 - x_2 - \cdots - x_{n-1}).$$

Note that

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ -x_1 - x_2 - \cdots - x_{n-1} \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ -1 \end{bmatrix} + \cdots + x_{n-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ -1 \end{bmatrix}$$

and so the vectors $\mathbf{u}_1 = (1, 0, \ldots, 0, -1), \mathbf{u}_2 = (0, 1, \ldots, 0, -1), \ldots, \mathbf{u}_{n-1} = (0, 0, \ldots, 1, -1)$ span W_1. If

$$\alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 + \cdots + \alpha_{n-1} \mathbf{u}_{n-1} = \mathbf{0},$$

then $(\alpha_1, \alpha_2, \ldots, \alpha_{n-1}, -\alpha_1 - \alpha_2 - \cdots - \alpha_{n-1}) = (0, 0, \ldots, 0, 0)$ and so $\alpha_1 = \alpha_2 = \cdots = \alpha_{n-1} = 0$. Hence $\mathbf{u}_1, \ldots, \mathbf{u}_{n-1}$ are linearly independent. So $\{\mathbf{u}_1, \ldots, \mathbf{u}_{n-1}\}$ is a basis for W_1 and $\dim(W_1) = n - 1$.

(b) Find a basis and state the dimension of W_2.
SOLUTION. Let $(x_1, x_2, \ldots, x_n) \in W_2$. Then

$$(x_1, x_2, \ldots, x_n) = (x_1, x_1, \ldots, x_1) = x_1(1, 1, \ldots, 1).$$

So the vector $\mathbf{u} = (1, 1, \ldots, 1)$ spans W_2. If $\alpha\mathbf{u} = \mathbf{0}$,then $\alpha = 0$ since $\mathbf{u} \neq \mathbf{0}$. Hence $\{\mathbf{u}\}$ is a basis for W_2 and $\dim(W_2) = 1$.

(c) Show that $\mathbb{R}^n = W_1 \oplus W_2$.
SOLUTION. Let $\mathbf{w} \in W_1 \cap W_2$. Then $\mathbf{w} \in W_2$ implies $\mathbf{w} = (\alpha, \alpha, \ldots, \alpha)$ but then $\mathbf{w} \in W_1$ implies that $\alpha + \alpha + \cdots + \alpha = 0$, so $n\alpha = 0$, so $\alpha = 0$, and so $\mathbf{w} = \mathbf{0}$. By Homework **4**, Problem **4**(a),

$$\dim(W_1 + W_2) + \dim(W_1 \cap W_2) = \dim(W_1) + \dim(W_2) = n - 1 + 1 = n.$$

Since $\dim(W_1 \cap W_2) = \dim\{\mathbf{0}\} = 0$, we have that $\dim(W_1 + W_2) = n = \dim(\mathbb{R}^n)$. Since $W_1 + W_2$ is a subspace of $\mathbb{R}^n$, we get $W_1 + W_2 = \mathbb{R}^n$. [Note: This statement is not true in general. For example, if $\mathbb{F} = \mathbb{Z}/2\mathbb{Z}$, ie. the field of two elements $\{0, 1\}$ under binary arithmetic, then $\mathbb{F}^2 \neq W_1 \oplus W_2$ since $W_1 \cap W_2 \neq \{\mathbf{0}_{\mathbb{F}^2}\}$. Why? Because $(1, 1) \in W_1$ (recall that $1 + 1 = 0$ in $\mathbb{F}$) and $(1, 1) \in W_2$ but $\mathbf{0}_{\mathbb{F}^2} = (0, 0) \neq (1, 1)$.]

3. Let $\alpha, \beta, \gamma, \delta \in \mathbb{R}$. Which of the following functions $T_i : \mathbb{R}^2 \to \mathbb{R}^2$ $(i = a, b, c, d)$ are linear? Justify your answers.
(a) $T_a(x, y) = (\alpha x + \beta y, \gamma x + \delta y)$,
(b) $T_b(x, y) = (\alpha x^2 + \beta y^2, \gamma x^2 + \delta y^2)$,
(c) $T_c(x, y) = (\alpha^2 x + \beta^2 y, \gamma^2 x + \delta^2 y)$,
(d) $T_d(x, y) = (\alpha x + \beta, \gamma + \delta y)$,
where $(x, y) \in \mathbb{R}^2$.
SOLUTION. T_a and T_c are linear. Let $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ and $\lambda, \mu \in \mathbb{R}$. Then

$$\begin{aligned} T_a(\lambda(x_1, y_1) + \mu(x_2, y_2)) &= T_a(\lambda x_1 + \mu x_2, \lambda y_1 + \mu y_2) \\ &= (\alpha(\lambda x_1 + \mu x_2) + \beta(\lambda y_1 + \mu y_2), \gamma(\lambda x_1 + \mu x_2) + \delta(\lambda y_1 + \mu y_2)) \\ &= (\alpha(\lambda x_1 + \mu x_2) + \beta(\lambda y_1 + \mu y_2), \gamma(\lambda x_1 + \mu x_2) + \delta(\lambda y_1 + \mu y_2)) \\ &= (\alpha\lambda x_1 + \alpha\mu x_2 + \beta\lambda y_1 + \beta\mu y_2, \gamma\lambda x_1 + \gamma\mu x_2 + \delta\lambda y_1 + \delta\mu y_2) \\ &= (\lambda\alpha x_1 + \lambda\beta y_1 + \mu\alpha x_2 + \mu\beta y_2, \lambda\gamma x_1 + \lambda\delta y_1 + \mu\gamma x_2 + \mu\delta y_2) \\ &= (\lambda(\alpha x_1 + \beta y_1) + \mu(\alpha x_2 + \beta y_2), \lambda(\gamma x_1 + \delta y_1) + \mu(\gamma x_2 + \delta y_2)) \\ &= \lambda(\alpha x_1 + \beta y_1, \gamma x_1 + \delta y_1) + \mu(\alpha x_2 + \beta y_2, \gamma x_2 + \delta y_2) \\ &= \lambda T_a(x_1, y_1) + \mu T_a(x_2, y_2). \end{aligned}$$

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小红书号:3189205503 qq

T_c is just T_a with $\alpha, \beta, \gamma, \delta$ (which can be any constants in $\mathbb{R}$) being $\alpha^2, \beta^2, \gamma^2, \delta^2$. T_b is not linear unless $\alpha = \beta = \gamma = \delta = 0$. Say, suppose $\gamma \neq 0$, then

$$T_b(-(1,0)) = T_b(-1,0) = (\alpha, \gamma) \neq (-\alpha, -\gamma) = -T_b(1,0)$$

since $\gamma \neq -\gamma$. T_d is not linear unless $\beta = \gamma = 0$ since

$$T_d(0,0) = (\beta, \gamma) \neq (0,0)$$

if $\beta \neq 0$ or $\gamma \neq 0$.

4. Let $\mathbb{P}$ be the vector space of polynomials with coefficients in $\mathbb{R}$. Which of the following functions $S_i : \mathbb{P} \to \mathbb{P}$ $(i = a, b, c, d)$ are linear? Justify your answers.
(a) $S_a(p(x)) = p(x^2)$,
(b) $S_b(p(x)) = [p(x)]^2$,
(c) $S_c(p(x)) = x^2 p(x)$,
(d) $S_d(p(x)) = p(x+2)$,
where $p(x) \in \mathbb{P}$.

SOLUTION. S_a, S_c, and S_d are linear. Let $p(x), q(x) \in \mathbb{P}$ and $\lambda, \mu \in \mathbb{R}$. Let

$$d = \max\{\deg(p(x)), \deg(q(x))\}.$$

Then we may write[1]

$$p(x) = a_0 + a_1 x + \cdots + a_d x^d \qquad \text{and} \qquad q(x) = b_0 + b_1 x + \cdots + b_d x^d.$$

S_a is linear because

$$\begin{aligned}
S_a(\lambda p(x) + \mu q(x)) &= S_a(\lambda(a_0 + a_1 x + a_2 x^2 + \cdots + a_d x^d) + \mu(b_0 + b_1 x + b_2 x^2 + \cdots + b_d x^d)) \\
&= S_a((\lambda a_0 + \mu b_0) + (\lambda a_1 + \mu b_1)x + (\lambda a_2 + \mu b_2)x^2 + \cdots + (\lambda a_d + \mu b_d)x^d) \\
&= (\lambda a_0 + \mu b_0) + (\lambda a_1 + \mu b_1)x^2 + (\lambda a_2 + \mu b_2)x^4 + \cdots + (\lambda a_d + \mu b_d)x^{2d} \\
&= \lambda(a_0 + a_1 x^2 + a_2 x^4 + \cdots + a_d x^{2d}) + \mu(b_0 + b_1 x^2 + b_2 x^4 + \cdots + b_d x^{2d}) \\
&= \lambda p(x^2) + \mu q(x^2) \\
&= \lambda S_a(p(x)) + \mu S_a(q(x)).
\end{aligned}$$

S_c is linear because

$$\begin{aligned}
S_c(\lambda p(x) + \mu q(x)) &= S_c(\lambda(a_0 + a_1 x + a_2 x^2 + \cdots + a_d x^d) + \mu(b_0 + b_1 x + b_2 x^2 + \cdots + b_d x^d)) \\
&= S_c((\lambda a_0 + \mu b_0) + (\lambda a_1 + \mu b_1)x + (\lambda a_2 + \mu b_2)x^2 + \cdots + (\lambda a_d + \mu b_d)x^d) \\
&= x^2[(\lambda a_0 + \mu b_0) + (\lambda a_1 + \mu b_1)x + (\lambda a_2 + \mu b_2)x^2 + \cdots + (\lambda a_d + \mu b_d)x^d] \\
&= (\lambda a_0 + \mu b_0)x^2 + (\lambda a_1 + \mu b_1)x^3 + (\lambda a_2 + \mu b_2)x^4 + \cdots + (\lambda a_d + \mu b_d)x^{d+2} \\
&= \lambda x^2(a_0 + a_1 x + a_2 x^2 + \cdots + a_d x^d) + \mu x^2(b_0 + b_1 x + b_2 x^2 + \cdots + b_d x^d) \\
&= \lambda x^2 p(x) + \mu x^2 q(x) \\
&= \lambda S_c(p(x)) + \mu S_a(q(x)).
\end{aligned}$$

S_d is linear because

$$\begin{aligned}
S_d(\lambda p(x) + \mu q(x)) &= S_d(\lambda(a_0 + a_1 x + a_2 x^2 + \cdots + a_d x^d) + \mu(b_0 + b_1 x + b_2 x^2 + \cdots + b_d x^d)) \\
&= S_d((\lambda a_0 + \mu b_0) + (\lambda a_1 + \mu b_1)x + (\lambda a_2 + \mu b_2)x^2 + \cdots + (\lambda a_d + \mu b_d)x^d) \\
&= (\lambda a_0 + \mu b_0) + (\lambda a_1 + \mu b_1)(x+2) + (\lambda a_2 + \mu b_2)(x+2)^2 + \cdots + (\lambda a_d + \mu b_d)(x+2)^d \\
&= \lambda(a_0 + a_1(x+2) + a_2(x+2)^2 + \cdots + a_d(x+2)^d) \\
&\qquad + \mu(b_0 + b_1(x+2) + b_2(x+2)^2 + \cdots + b_d(x+2)^d) \\
&= \lambda p(x+2) + \mu q(x+2) \\
&= \lambda S_d(p(x)) + \mu S_d(q(x)).
\end{aligned}$$

S_b is not linear since

$$S_b(-x) = (-x)^2 = x^2 \neq -x^2 = -S_c(x).$$

[1] If, for example, $\deg(p) > \deg(q)$, then $d = \deg(p)$ and it is understood that $b_{\deg(q)+1} = \cdots = b_d = 0$.

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5. We have shown that $\mathbb{R}_+ = \{x \in \mathbb{R} \mid x > 0\}$ is a vector space over $\mathbb{R}$ with addition and scalar multiplication defined by

$$x \oplus y = xy \qquad \text{and} \qquad \lambda \odot x = x^\lambda$$

for $x, y \in \mathbb{R}_+$, $\lambda \in \mathbb{R}$. Show that $\log_2 : \mathbb{R}_+ \to \mathbb{R}$ is a linear transformation. What about $\log_b : \mathbb{R}_+ \to \mathbb{R}$ for other bases $b > 2$?

SOLUTION. Let $\alpha, \beta \in \mathbb{R}$ and $x, y \in \mathbb{R}_+$. Then

$$\begin{aligned}
\log_2(\alpha \odot x \oplus \beta \odot y) &= \log_2(x^\alpha \oplus y^\beta) \\
&= \log_2(x^\alpha y^\beta) \\
&= \log_2(x^\alpha) + \log_2(y^\beta) \\
&= \alpha \log_2(x) + \beta \log_2(y)
\end{aligned}$$

by the basic properties of logarithms. Hence $\log_2 : \mathbb{R}_+ \to \mathbb{R}$ is a linear transformation. Again using a basic property of logarithms, namely,

$$\log_b(x) = \frac{\log_2(x)}{\log_2(b)},$$

we see that

$$\log_b = \gamma \log_2$$

where $\gamma = 1/\log_2(b) \in \mathbb{R}$. By a result in the lectures, any scalar multiple of a linear transformation must also be a linear transformation. So $\log_b : \mathbb{R}_+ \to \mathbb{R}$ is a linear transformation.

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MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 6 SOLUTIONS

You may use without proof any results from the lectures as well as any basic properties of matrix operations and calculus operations. $\mathbb{P}$ will denote the vector space of polynomials in x with coefficients in $\mathbb{R}$.

1. Let V be a vector space over $\mathbb{R}$. Show that the following maps $T_i : V \to V$, $i = a, b, c, d$, are linear operators.

(a) $V = \mathbb{R}^{m\times m}$, $A \in \mathbb{R}^{m\times m}$ is a fixed matrix, and $T_a : \mathbb{R}^{m\times m} \to \mathbb{R}^{m\times m}$ is defined by

$$T_a(X) = AX - XA$$

for all $X \in \mathbb{R}^{m\times m}$.

SOLUTION. Let $\alpha, \beta \in \mathbb{R}$ and $X_1, X_2 \in \mathbb{R}^{m\times m}$. Then

$$\begin{aligned} T_a(\alpha X_1 + \beta X_2) &= A(\alpha X_1 + \beta X_2) - (\alpha X_1 + \beta X_2)A \\ &= \alpha AX_1 + \beta AX_2 - \alpha X_1 A - \beta X_2 A \\ &= \alpha(AX_1 - X_1A) + \beta(AX_2 - X_2A) \\ &= \alpha T_a(X_1) + \beta T_b(X_2). \end{aligned}$$

(b) $V = \mathbb{R}^{m\times n}$, $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{n\times n}$ are fixed matrices, and $T_b : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$ is defined by

$$T_b(X) = AXB$$

for all $X \in \mathbb{R}^{m\times n}$.

SOLUTION. Let $\alpha, \beta \in \mathbb{R}$ and $X_1, X_2 \in \mathbb{R}^{m\times m}$. Then

$$\begin{aligned} T_b(\alpha X_1 + \beta X_2) &= A(\alpha X_1 + \beta X_2)B \\ &= \alpha AX_1B + \beta AX_2B \\ &= \alpha T_b(X_1) + \beta T_b(X_2). \end{aligned}$$

(c) $V = \mathbb{P}$, and $T_c : \mathbb{P} \to \mathbb{P}$ is defined by

$$T_c(p(x)) = p'(x)$$

for all $p(x) \in \mathbb{P}$.

SOLUTION. Let $\alpha, \beta \in \mathbb{R}$ and $p(x), q(x) \in \mathbb{P}$. Then by the basic properties of differentiation that you have learned in high-school calculus,

$$\begin{aligned} T_c(\alpha p(x) + \beta q(x)) &= \frac{d}{dx}(\alpha p(x) + \beta q(x)) \\ &= \alpha \frac{d}{dx} p(x) + \beta \frac{d}{dx} q(x) \\ &= \alpha p'(x) + \beta q'(x) \\ &= \alpha T_c(p(x)) + \beta T_c(q(x)). \end{aligned}$$

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小红书
小红书号:3189205503 qq

(d) $V = \mathbb{P}$, and $T_d : \mathbb{P} \to \mathbb{P}$ is defined by

$$T_d(p(x)) = xp(x)$$

for all $p(x) \in \mathbb{P}$.

SOLUTION. Let $\alpha, \beta \in \mathbb{R}$ and $p(x), q(x) \in \mathbb{P}$. Then by the basic properties of polynomial multiplication that you have learned in high-school algebra,

$$\begin{aligned} T_d(\alpha p(x) + \beta q(x)) &= x(\alpha p(x) + \beta q(x)) \\ &= \alpha x p(x) + \beta x q(x) \\ &= \alpha T_d(p(x)) + \beta T_d(q(x)). \end{aligned}$$

2. Let $F : \mathbb{R}^3 \to \mathbb{R}^2$, $G : \mathbb{R}^3 \to \mathbb{R}^2$, $H : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $F(x, y, z) = (y, x + z)$, $G(x, y, z) = (2z, x - y)$, $H(x, y) = (y, 2x)$ respectively.

(a) Find the formulas that define $3F - 5G$, $H \circ F$, $H \circ G$, $H \circ (F + G)$, and state the respective domains and codomains of these maps. Are these linear transformations?

SOLUTION. $3F - 5G : \mathbb{R}^3 \to \mathbb{R}^2$ and its formula is given by

$$\begin{aligned} (3F - 5G)(x, y, z) &= 3F(x, y, z) - 5G(x, y, z) \\ &= 3(y, x + z) - 5(2z, x - y) \\ &= (3y - 10z, -2x + 3z + 5y). \end{aligned}$$

$H \circ F : \mathbb{R}^3 \to \mathbb{R}^2$ and its formula is given by

$$H \circ F(x, y, z) = H(F(x, y, z)) = H(y, x + z) = (x + z, 2y).$$

$H \circ G : \mathbb{R}^3 \to \mathbb{R}^2$ and its formula is given by

$$H \circ G(x, y, z) = H(G(x, y, z)) = H(2z, x - y) = (x - y, 4z).$$

$H \circ (F + G) : \mathbb{R}^3 \to \mathbb{R}^2$ and its formula is given by

$$\begin{aligned} H \circ (F + G)(x, y, z) &= H((F + G)(x, y, z)) \\ &= H(y + 2z, 2x - y + z) \\ &= (2x - y + z, 2y + 4z). \end{aligned}$$

Alternatively, we may use Problem **3**(c) below: $H \circ (F + G) = H \circ F + H \circ G$. All four maps are linear transformations.

(b) We define $H^2 = H \circ H$, $H^3 = H \circ H \circ H$, and so on. Find the formulas that define H^2, H^3, and $H^2 \circ (F + G)$, and state the respective domains and codomains of these maps. Are these linear transformations?

SOLUTION. $H^2 : \mathbb{R}^2 \to \mathbb{R}^2$ and its formula is given by

$$H^2(x, y) = H(H(x, y)) = H(y, 2x) = (2x, 2y).$$

$H^2 : \mathbb{R}^2 \to \mathbb{R}^2$ and its formula is given by

$$H^3(x, y) = H(H^2(x, y)) = H(2x, 2y) = (2y, 4x).$$

$H^2 \circ (F + G) : \mathbb{R}^3 \to \mathbb{R}^2$ and its formula is given by

$$\begin{aligned} H^2 \circ (F + G)(x, y) &= H^2((F + G)(x, y)) \\ &= H^2(y + 2z, 2x - y + z) \\ &= (2y + 4z, 4x - 2y + 2z). \end{aligned}$$

All three maps are linear transformations.

3. Let U, W, V be finite-dimensional vector spaces over $\mathbb{F}$. Let $\alpha, \beta \in \mathbb{F}$.

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小红书号:3189205503 qq

(a) Let $T : V \to W$ be a linear transformation. Show that

$$\operatorname{rank}(T) \leq \dim(V).$$

SOLUTION. Let $\dim(V) = n$. Suppose $\dim(\operatorname{Im}(T)) = \operatorname{rank}(T) > n$. Then there must be at least $n+1$ vectors $\mathbf{w}_1, \ldots, \mathbf{w}_{n+1} \in \operatorname{Im}(T)$ that are linearly independent (eg. choose the first $n+1$ vectors in a basis of $\operatorname{Im}(T)$). Since $\mathbf{w}_1, \ldots, \mathbf{w}_{n+1}$ are in $\operatorname{Im}(T)$, there exist $\mathbf{v}_1, \ldots, \mathbf{v}_{n+1} \in V$ such that $T(\mathbf{v}_i) = \mathbf{w}_i$, $i = 1, \ldots, n+1$. Since $\dim(V) = n$, $\mathbf{v}_1, \ldots, \mathbf{v}_{n+1}$ must be linearly dependent, so there exist $\alpha_1, \ldots, \alpha_{n+1} \in \mathbb{F}$, not all zero, such that

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_{n+1} \mathbf{v}_{n+1} = \mathbf{0}_V.$$

Hence we have found $\alpha_1, \ldots, \alpha_{n+1} \in \mathbb{F}$, not all zero, such that

$$T(\alpha_1 \mathbf{v}_1 + \cdots + \alpha_{n+1} \mathbf{v}_{n+1}) = T(\mathbf{0}_V),$$

ie.

$$\alpha_1 T(\mathbf{v}_1) + \cdots + \alpha_{n+1} T(\mathbf{v}_{n+1}) = \mathbf{0}_W,$$

ie.

$$\alpha_1 \mathbf{w}_1 + \cdots + \alpha_{n+1} \mathbf{w}_{n+1} = \mathbf{0}_W,$$

which implies that $\mathbf{w}_1, \ldots, \mathbf{w}_{n+1}$ are linearly dependent — a contradiction. Hence our original assumption must have been false, ie. we must have $\dim(\operatorname{Im}(T)) \leq n$.

(b) Let $S : U \to V$ and $T : V \to W$ be linear transformations. Show that

$$\operatorname{rank}(T \circ S) \leq \operatorname{rank}(T)$$

and

$$\operatorname{rank}(T \circ S) \leq \operatorname{rank}(S).$$

SOLUTION. Recall the following elementary fact from set theory (Math **74**): $A \subseteq B$ implies $f(A) \subseteq f(B)$ for any function f. In particular, since $S(U) \subseteq V$, we must have $T(S(U)) \subseteq T(V)$, ie. $T \circ S(U) \subseteq T(V)$, ie.

$$\operatorname{im}(T \circ S) \subseteq \operatorname{im}(T)$$

(recall that $\operatorname{im}(\varphi)$ is just another way of writing $\varphi(V)$ — both denote the range of φ). Now both of these are subspaces of W and so

$$\dim(\operatorname{im}(T \circ S)) \leq \dim(\operatorname{im}(T)),$$

ie.

$$\operatorname{rank}(T \circ S) \leq \operatorname{rank}(T).$$

For the second part, we define the function $\varphi : \operatorname{im}(S) \to W$ by $\varphi(\mathbf{v}) = T(\mathbf{v})$ for all $\mathbf{v} \in \operatorname{im}(S)$. Note that φ is a linear transformation since for all $\mathbf{v}_1, \mathbf{v}_2 \in \operatorname{im}(S)$ and $\alpha, \beta \in \mathbb{F}$, we have $\varphi(\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) = T(\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) = \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2) = \alpha \varphi(\mathbf{v}_1) + \beta \varphi(\mathbf{v}_2)$. So applying part (a) to φ, we get

$$\operatorname{rank}(\varphi) \leq \dim(\operatorname{im}(S)) = \operatorname{rank}(S). \tag{3.1}$$

But note that $\operatorname{im}(T \circ S) \subseteq \operatorname{im}(\varphi)$ — since if $\mathbf{w} \in \operatorname{im}(T \circ S)$, then $\mathbf{w} = T(S(\mathbf{u}))$ for some $\mathbf{u} \in U$ and therefore $\mathbf{w} = T(\mathbf{v}) = \varphi(\mathbf{v})$ where $\mathbf{v} = S(\mathbf{u}) \in \operatorname{im}(S)$; hence $\mathbf{w} \in \operatorname{im}(\varphi)$. So

$$\dim(\operatorname{im}(T \circ S)) \leq \dim(\operatorname{im}(\varphi)),$$

ie.

$$\operatorname{rank}(T \circ S) \leq \operatorname{rank}(\varphi). \tag{3.2}$$

Combining (3.1) and (3.2) then yields

$$\operatorname{rank}(T \circ S) \leq \operatorname{rank}(S).$$

小红书
小红书号:3189205503 qq

(c) Let $S_1 : U \to V$, $S_2 : U \to V$, and $T : V \to W$ be linear transformations. Show that

$$T \circ (\alpha S_1 + \beta S_2) = \alpha T \circ S_1 + \beta T \circ S_2.$$

SOLUTION. Let $\mathbf{u} \in U$. Then

$$\begin{aligned} T \circ (\alpha S_1 + \beta S_2)(\mathbf{u}) &= T((\alpha S_1 + \beta S_2)(\mathbf{u})) \\ &= T(\alpha S_1(\mathbf{u}) + \beta S_2(\mathbf{u})) \\ &= \alpha T(S_1(\mathbf{u})) + \beta T(S_2(\mathbf{u})) \\ &= \alpha T \circ S_1(\mathbf{u}) + \beta T \circ S_2(\mathbf{u}). \end{aligned}$$

Since this is true for any $\mathbf{u} \in U$, we have that

$$T \circ (\alpha S_1 + \beta S_2) = \alpha T \circ S_1 + \beta T \circ S_2.$$

(d) Let $S : U \to V$, $T_1 : V \to W$, and $T_2 : V \to W$ be linear transformations. Show that

$$(\alpha T_1 + \beta T_2) \circ S = \alpha T_1 \circ S + \beta T_2 \circ S.$$

SOLUTION. Let $\mathbf{u} \in U$. Then

$$\begin{aligned} (\alpha T_1 + \beta T_2) \circ S(\mathbf{u}) &= (\alpha T_1 + \beta T_2)(S(\mathbf{u})) \\ &= \alpha T_1(S(\mathbf{u})) + \beta T_2(S(\mathbf{u})) \\ &= \alpha T_1 \circ S(\mathbf{u}) + \beta T_2 \circ S(\mathbf{u}) \end{aligned}$$

Since this is true for any $\mathbf{u} \in U$, we have that

$$(\alpha T_1 + \beta T_2) \circ S = \alpha T_1 \circ S + \beta T_2 \circ S.$$

4. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation such that

$$T\left(\begin{bmatrix}1\\0\\1\end{bmatrix}\right) = \begin{bmatrix}2\\3\\-1\end{bmatrix}, \quad T\left(\begin{bmatrix}1\\-1\\1\end{bmatrix}\right) = \begin{bmatrix}3\\0\\-2\end{bmatrix}, \quad T\left(\begin{bmatrix}-2\\7\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\3\\-1\end{bmatrix}.$$

Find $\mathrm{im}(T)$, $\ker(T)$, and the formula that defines T.

SOLUTION. Let

$$\mathbf{u}_1 = \begin{bmatrix}1\\0\\1\end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix}1\\-1\\1\end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix}-2\\7\\-1\end{bmatrix}.$$

Now if $\alpha\mathbf{u}_1 + \beta\mathbf{u}_2 + \gamma\mathbf{u}_3 = \mathbf{0}$, then solving the system gives $\alpha = \beta = \gamma = 0$. So $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are linearly independent and since $\dim(\mathbb{R}^3) = 3$, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a basis for $\mathbb{R}^3$. Hence by a result in the lectures, we have

$$\mathrm{im}(T) = \mathrm{span}\{T(\mathbf{u}_1), T(\mathbf{u}_2), T(\mathbf{u}_3)\}.$$

We will use the Throwing-Out Algorithm to obtain a basis for $\mathrm{im}(T)$. Since $T(\mathbf{u}_1) = T(\mathbf{u}_3)$, we may throw out $T(\mathbf{u}_3)$. The remain set $\{T(\mathbf{u}_1), T(\mathbf{u}_2)\}$ is easily seen to be linearly independent:

$$\lambda \begin{bmatrix}2\\3\\-1\end{bmatrix} + \mu \begin{bmatrix}3\\0\\-2\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

implies $\lambda = 0$ (just examine the second coordinate) and so $\mu = 0$. Hence $\{T(\mathbf{u}_1), T(\mathbf{u}_2)\}$ is a basis and so

$$\mathrm{im}(T) = \mathrm{span}\left\{\begin{bmatrix}2\\3\\-1\end{bmatrix}, \begin{bmatrix}3\\0\\-2\end{bmatrix}\right\}.$$

小红书
小红书号:3189205503 qq

To find the formula defines T we will first need to determine how an arbitrary $\mathbf{u} = [x, y, z]^\top \in \mathbb{R}^3$ may be expressed in terms of the basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. So we let

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} -2 \\ 7 \\ -1 \end{bmatrix}$$

and solve the system to get

$$\alpha = 6x + y - 5z, \quad \beta = -7x - y + 7z, \quad \gamma = -x + z. \tag{4.3}$$

Since T is a linear operator,

$$\begin{aligned} T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) &= \alpha T\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) + \beta T\left(\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right) + \gamma T\left(\begin{bmatrix} -2 \\ 7 \\ -1 \end{bmatrix}\right) \\ &= \alpha \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} + \beta \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix} + \gamma \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 2(\alpha+\gamma) + 3\beta \\ 3(\alpha+\gamma) \\ -(\alpha+\gamma) - 2\beta \end{bmatrix} \\ &= \begin{bmatrix} -11x - y + 13z \\ 15x + 3y - 12z \\ 9x + y - 10z \end{bmatrix} \end{aligned}$$

upon substituting (4.3). To find $\ker(T)$, we set $T(\mathbf{u}) = \mathbf{0}$ and using the next to last expression in (4.3), we get

$$\begin{bmatrix} 2(\alpha+\gamma) + 3\beta \\ 3(\alpha+\gamma) \\ -(\alpha+\gamma) - 2\beta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

which gives us $\alpha = -\gamma$ and $\beta = 0$; ie. $6x + y - 5z = x - z$ and $-7x - y + 7z = 0$, ie. $x = \frac{3}{2}z$ and $y = -\frac{7}{2}z$. Hence

$$\ker(T) = \operatorname{span}\left\{\begin{bmatrix} 3 \\ -7 \\ 2 \end{bmatrix}\right\}.$$

5. Let $T : V \to W$ be a linear transformation. Let V be finite-dimensional and $\dim(V) = n$. If $\mathbf{w}_1, \ldots, \mathbf{w}_m \in W$ form a basis for $\operatorname{im}(T)$ and if $\mathbf{v}_1, \ldots, \mathbf{v}_m \in V$ are such that

$$T(\mathbf{v}_i) = \mathbf{w}_i$$

for $i = 1, \ldots, m$, show that

$$V = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} \oplus \ker(T).$$

SOLUTION. Let $\mathbf{v} \in V$. Then $T(\mathbf{v}) \in \operatorname{Im}(T)$. Since $\mathbf{w}_1, \ldots, \mathbf{w}_m$ form a basis for $\operatorname{im}(T)$, there exist $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$T(\mathbf{v}) = \alpha_1 \mathbf{w}_1 + \cdots + \alpha_m \mathbf{w}_m.$$

Since $T(\mathbf{v}_i) = \mathbf{w}_i$, we get

$$T(\mathbf{v}) = \alpha_1 T(\mathbf{v}_1) + \cdots + \alpha_m T(\mathbf{v}_m),$$

ie.

$$T(\mathbf{v}) - \alpha_1 T(\mathbf{v}_1) - \cdots - \alpha_m T(\mathbf{v}_m) = \mathbf{0}_W,$$

ie.

$$T(\mathbf{v} - \alpha_1 \mathbf{v}_1 - \cdots - \alpha_m \mathbf{v}_m) = \mathbf{0}_W$$

小红书
小红书号:3189205503 qq

which implies that $\mathbf{v} - \alpha_1\mathbf{v}_1 - \cdots - \alpha_m\mathbf{v}_m \in \ker(T)$. So given any $\mathbf{v} \in V$, we could write

$$\mathbf{v} = (\alpha_1\mathbf{v}_1 + \cdots + \alpha_m\mathbf{v}_m) + (\mathbf{v} - \alpha_1\mathbf{v}_1 - \cdots - \alpha_m\mathbf{v}_m) \in \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} + \ker(T),$$

which means that

$$V = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} + \ker(T).$$

We will now show that this sum is in fact a direct sum. Let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be a basis for $\ker(T)$ where $k = \text{nullity}(T)$. We claim that $\{\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is linearly independent and by Homework **3**, Problem **4**(b),

$$\begin{aligned} V &= \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} + \ker(T) \\ &= \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} + \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \\ &= \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{u}_1, \ldots, \mathbf{u}_k\} \\ &= \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} \oplus \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \\ &= \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} \oplus \ker(T). \end{aligned}$$

Suppose there exists $\lambda_1, \ldots, \lambda_m, \mu_1, \ldots, \mu_k \in \mathbb{F}$ such that

$$\lambda_1\mathbf{v}_1 + \cdots + \lambda_m\mathbf{v}_m + \mu_1\mathbf{u}_1 + \cdots + \mu_k\mathbf{u}_k = \mathbf{0}_V. \tag{5.4}$$

Then applying T to both sides of the equation gives

$$\lambda_1 T(\mathbf{v}_1) + \cdots + \lambda_m T(\mathbf{v}_m) + \mu_1 T(\mathbf{u}_1) + \cdots + \mu_k T(\mathbf{u}_k) = \mathbf{0}_W.$$

Now since $T(\mathbf{u}_i) = \mathbf{0}_W$ for all $i = 1, \ldots, k$, and $T(\mathbf{v}_j) = \mathbf{w}_j$ for all $j = 1, \ldots, m$, we get

$$\lambda_1\mathbf{w}_1 + \cdots + \lambda_m\mathbf{w}_m = \mathbf{0}_W.$$

As $\mathbf{w}_1, \ldots, \mathbf{w}_m$ are linearly independent, we get

$$\lambda_1 = \cdots = \lambda_m = 0.$$

So we are left with

$$\mu_1\mathbf{u}_1 + \cdots + \mu_k\mathbf{u}_k = \mathbf{0}_V$$

in (5.4). But since $\mathbf{u}_1, \ldots, \mathbf{u}_k$ are also linearly independent, we then get

$$\mu_1 = \cdots = \mu_k = 0.$$

So $\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{u}_1, \ldots, \mathbf{u}_k$ are linearly independent.

小红书
小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 7 SOLUTIONS

Let V be a vector space. The *identity transformation* on V is denoted by I_V, ie. $I_V : V \to V$ and $I_V(\mathbf{u}) = \mathbf{u}$ for all $\mathbf{u} \in V$. The *zero transformation* on V is denoted by O_V, ie. $O_V : V \to V$ and $O_V(\mathbf{u}) = \mathbf{0}_V$ for all $\mathbf{u} \in V$ where $\mathbf{0}_V$ is the zero vector/additive identity of V.

1. Find the kernels of the following linear transformation.

(a) $T_a : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T_a(x, y) = (2x + 3y, 7x - 5y)$$

for all $(x, y) \in \mathbb{R}^2$.
SOLUTION. $T_a(x, y) = (0, 0)$ iff $(2x + 3y, 7x - 5y) = (0, 0)$ iff $x = y = 0$. Hence $\ker(T_a) = \{(0, 0)\}$.

(b) $T_b : \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$T_b(x, y) = (x, 0, 0)$$

for all $(x, y) \in \mathbb{R}^2$.
SOLUTION. $T_b(x, y) = (0, 0, 0)$ iff $x = 0$. Hence $\ker(T_b) = \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\} = \text{span}\{(0, 1)\}$.

(c) $T_c : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$T_c(x, y) = x + 2y$$

for all $(x, y) \in \mathbb{R}^2$.
SOLUTION. $T_c(x, y) = 0$ iff $x = -2y$. Hence $\ker(T_c) = \{(-2y, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\} = \text{span}\{(-2, 1)\}$.

(d) $T_d : \mathbb{P}_5 \to \mathbb{P}_{10}$ defined by

$$T_d(p(x)) = p(x^2)$$

for all $p(x) \in \mathbb{P}_5$.
SOLUTION. Let $0_n = 0 + 0x + \cdots + 0x^n$. $T_d(p(x)) = 0_{10}$ iff $p(x^2) = 0_{10}$ iff $p(x) = 0_5$. Hence $\ker(T_d) = \{0_5\}$.

(e) $T_e : \mathbb{P}_5 \to \mathbb{P}_4$ defined by

$$T_e(p(x)) = p'(x)$$

for all $p(x) \in \mathbb{P}_5$.
SOLUTION. $T_e(p(x)) = 0_4$ iff $p'(x) = 0_4$ iff $p(x) = c$ (constant polynomial). Hence $\ker(T_e) = \{c\} = \text{span}\{1\}$.

(f) $T_f : \mathbb{P}_5 \to \mathbb{P}_6$ defined by

$$T_f(p(x)) = \int_{-3}^{x+9} p(t)\, dt$$

for all $p(x) \in \mathbb{P}_5$.
SOLUTION. $T_f(p(x)) = 0_6$ iff $\int_{-3}^{x+9} p(t)\, dt = 0_6$ iff $p(x) = 0_5$. Hence $\ker(T_f) = \{0_5\}$.

2. Let $T : U \to V$ and $S : V \to W$ be linear transformations.

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小红书
小红书号:3189205503 qq

(a) Show that

$$\operatorname{im}(S \circ T) \subseteq \operatorname{im}(S)$$

and

$$\ker(S \circ T) \supseteq \ker(T).$$

SOLUTION. Let $\mathbf{w} \in \operatorname{im}(S \circ T)$. Then there exist some $\mathbf{u} \in U$ such that $\mathbf{w} = S \circ T(\mathbf{u}) = S(T(\mathbf{u}))$. So $\mathbf{w} \in \operatorname{im}(S)$ since $\mathbf{w} = S(\mathbf{v})$ where $\mathbf{v} = T(\mathbf{u}) \in V$. Hence $\operatorname{im}(S \circ T) \subseteq \operatorname{im}(S)$. For the second part, let $\mathbf{u} \in \ker(T)$. Then $T(\mathbf{u}) = \mathbf{0}_V$ and so $S(T(\mathbf{u})) = S(\mathbf{0}_V)$, ie. $S \circ T(\mathbf{u}) = \mathbf{0}_W$, ie. $\mathbf{u} \in \ker(S \circ T)$. Hence $\ker(T) \subseteq \ker(S \circ T)$.

(b) Show that if T is surjective, then

$$\operatorname{im}(S \circ T) = \operatorname{im}(S).$$

SOLUTION. By part (a), we know that $\operatorname{im}(S \circ T) \subseteq \operatorname{im}(S)$. We will show the reverse inclusion. If $\mathbf{w} \in \operatorname{im}(S)$, then there exists some $\mathbf{v} \in V$ such that $\mathbf{w} = S(\mathbf{v})$. But since $T : U \to V$ is surjective, there exists some $\mathbf{u} \in U$ such that $\mathbf{v} = T(\mathbf{u})$. So $\mathbf{w} = S(T(\mathbf{u})) = S \circ T(\mathbf{u})$, ie. $\mathbf{w} \in \operatorname{im}(S \circ T)$. Hence $\operatorname{im}(S) \subseteq \operatorname{im}(S \circ T)$.

(c) Show that if S is injective, then

$$\ker(S \circ T) = \ker(T).$$

SOLUTION. By part (a), we know that $\ker(S \circ T) \supseteq \ker(T)$. We will show the reverse inclusion. If $\mathbf{u} \in \ker(S \circ T)$, then $S(T(\mathbf{u})) = S \circ T(\mathbf{u}) = \mathbf{0}_W$. But since S is injective (and linear), by a result in the lectures, we $\ker(S) = \{\mathbf{0}_V\}$. So $T(\mathbf{u}) \in \ker(S)$ implies that $T(\mathbf{u}) = \mathbf{0}_V$, which in turns implies that $\mathbf{u} \in \ker(T)$. Hence $\ker(S \circ T) \subseteq \ker(T)$.

3. Let V be a finite-dimensional vector space. Let $S : V \to V$ and $T : V \to V$ be linear operators on V.

(a) Show that

$$\operatorname{rank}(S \circ T) \leq \min\{\operatorname{rank}(S), \operatorname{rank}(T)\}$$

and

$$\operatorname{nullity}(S \circ T) \leq \operatorname{nullity}(S) + \operatorname{nullity}(T).$$

SOLUTION. By Homework **6**, Problem **3**(b), we have that

$$\operatorname{rank}(S \circ T) \leq \operatorname{rank}(S) \quad \text{and} \quad \operatorname{rank}(S \circ T) \leq \operatorname{rank}(T).$$

Therefore

$$\operatorname{rank}(S \circ T) \leq \min\{\operatorname{rank}(S), \operatorname{rank}(T)\}.$$

For the second part, consider the function $\varphi : \ker(S \circ T) \to V$ defined by $\varphi(\mathbf{v}) = T(\mathbf{v})$ for all $\mathbf{v} \in \ker(S \circ T)$ (note that φ and T are two different functions because they have different domains). Since $\varphi(\alpha\mathbf{u} + \beta\mathbf{v}) = T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}) = \alpha\varphi(\mathbf{u}) + \beta\varphi(\mathbf{v})$, the function φ is a linear transformation from the vector space $\ker(S \circ T)$ to the vector space V. Now applying the rank-nullity theorem in the lectures to φ, we get

$$\dim(\ker(S \circ T)) = \operatorname{nullity}(\varphi) + \operatorname{rank}(\varphi) = \dim(\ker(\varphi)) + \dim(\operatorname{im}(\varphi)). \tag{3.1}$$

If $\mathbf{w} \in \operatorname{im}(\varphi)$, then $\mathbf{w} = \varphi(\mathbf{v})$ for some $\mathbf{v} \in \ker(S \circ T)$ and

$$S(\mathbf{w}) = S(\varphi(\mathbf{v})) = S(T(\mathbf{v})) = S \circ T(\mathbf{v}) = \mathbf{0}$$

and so $\mathbf{w} \in \ker(S)$. Hence $\operatorname{im}(\varphi) \subseteq \ker(S)$ and so

$$\dim(\operatorname{im}(\varphi)) \leq \dim(\ker(S)) = \operatorname{nullity}(S). \tag{3.2}$$

If $\mathbf{v} \in \ker(\varphi)$, then

$$\mathbf{0} = \varphi(\mathbf{v}) = T(\mathbf{v})$$

and so $\mathbf{v} \in \ker(T)$. Hence $\ker(\varphi) \subseteq \ker(T)$ and so

$$\dim(\ker(\varphi)) \leq \dim(\ker(T)) = \operatorname{nullity}(T). \tag{3.3}$$

小红书
小红书号:3189205503 qq

Applying (3.2) and (3.3) to the RHS of (3.1), we obtain

$$\text{nullity}(S \circ T) \leq \text{nullity}(S) + \text{nullity}(T).$$

(b) Show that

$$\text{rank}(S + T) \leq \text{rank}(S) + \text{rank}(T).$$

SOLUTION. Let $\mathbf{w} \in \text{im}(S + T)$. Then $\mathbf{w} = (S + T)(\mathbf{v})$ for some $\mathbf{v} \in V$. Observe that

$$\mathbf{w} = (S + T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) \in \text{im}(S) + \text{im}(T).$$

Hence $\text{im}(S + T) \subseteq \text{im}(S) + \text{im}(T)$ and hence

$$\dim(\text{im}(S + T)) \leq \dim(\text{im}(S) + \text{im}(T)) \leq \dim(\text{im}(S)) + \dim(\text{im}(T)),$$

where the second inequality follows from Homework **4**, Problem **4**(a). In other words, we have

$$\text{rank}(S + T) \leq \text{rank}(S) + \text{rank}(T).$$

(c) Show that if $S \circ T = O_V$, then

$$\text{rank}(S) + \text{rank}(T) \leq \dim(V).$$

SOLUTION. Note that $\ker(O_V) = V$ — for any $\mathbf{v} \in V$, $O_V(\mathbf{v}) = \mathbf{0}$, so $V \subseteq \ker(O_V)$; on the other hand, we always have $\ker(O_V) \subseteq V$. Hence $\text{nullity}(O_V) = \dim(V)$. Now we will apply part (a),

$$\text{nullity}(S \circ T) \leq \text{nullity}(S) + \text{nullity}(T)$$

to get

$$\dim(V) \leq \text{nullity}(S) + \text{nullity}(T).$$

Adding $\dim(V)$ to both sides of the inequality and bringing the two terms on the RHS to the LHS, we get

$$\dim(V) - \text{nullity}(S) + \dim(V) - \text{nullity}(T) \leq \dim(V).$$

Finally, we apply the rank-nullity theorem twice to get

$$\text{rank}(S) + \text{rank}(T) \leq \dim(V).$$

4. Let V be a finite-dimensional vector space. Let $T : V \to V$ be a linear operator on V. Show that the following three statements are equivalent:

(i) $V = \ker(T) + \text{im}(T)$;
(ii) $\ker(T) \cap \text{im}(T) = \{\mathbf{0}_V\}$;
(iii) $V = \ker(T) \oplus \text{im}(T)$.

SOLUTION. By Homework **4**, Problem **4**(a),

$$\begin{aligned}\dim(\ker(T) + \text{im}(T)) + \dim(\ker(T) \cap \text{im}(T)) &= \dim(\ker(T)) + \dim(\text{im}(T)) \\ &= \text{nullity}(T) + \text{rank}(T) \\ &= \dim(V).\end{aligned}$$

(a) $\Rightarrow$ (b): if $\ker(T) + \text{im}(T) = V$, then we get

$$\dim(V) + \dim(\ker(T) \cap \text{im}(T)) = \dim(V),$$

ie.

$$\dim(\ker(T) \cap \text{im}(T)) = 0,$$

ie. $\ker(T) \cap \text{im}(T) = \{\mathbf{0}_V\}$.

(b) $\Rightarrow$ (c): if $\ker(T) \cap \text{im}(T) = \{\mathbf{0}_V\}$, then we get

$$\dim(\ker(T) + \text{im}(T)) + \dim(\{\mathbf{0}_V\}) = \dim(V)$$

ie.

$$\dim(\ker(T) + \text{im}(T)) = \dim(V).$$

小红书
小红书号:3189205503 qq

Since $\ker(T) + \operatorname{im}(T)$ is a subspace of V, the only way their dimensions could be equal is if

$$\ker(T) + \operatorname{im}(T) = V.$$

But since $\ker(T) \cap \operatorname{im}(T) = \{\mathbf{0}_V\}$, the sum is a direct sum, ie.

$$\ker(T) \oplus \operatorname{im}(T) = V.$$

(c) $\Rightarrow$ (a): $\ker(T) \oplus \operatorname{im}(T) = V$ implies both (a) and (b), so it implies (a) in particular.

5. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (2x, 4x - y, 2x + 3y - z)$.

(a) Show that T is invertible.
SOLUTION. Let $(x, y, z) \in \ker(T)$. Then $T(x, y, z) = (2x, 4x - y, 2x + 3y - z) = (0, 0, 0)$. Solving this, we get $x = y = z = 0$ and so $(x, y, z) = (0, 0, 0)$ and so $\ker(T) = \{(0, 0, 0)\}$. Hence T is invertible by a result in the lectures.

(b) Find a formula for T^{-1}.
SOLUTION. Set $T(x, y, z) = (r, s, t)$ and so $T^{-1}(r, s, t) = (x, y, z)$. Now to find a formula for T^{-1} we solve $(2x, 4x - y, 2x + 3y - z) = (r, s, t)$ for x, y, z to get $(x, y, z) = (\frac{1}{2}r, 2r - s, 7r - 3s - t)$. Hence $T^{-1}(r, s, t) = (\frac{1}{2}r, 2r - s, 7r - 3s - t)$.

(c) Find a formula for $T^{-2} := T^{-1} \circ T^{-1}$.
SOLUTION. $T^{-2}(r, s, t) = T^{-1}(\frac{1}{2}r, 2r - s, 7r - 3s - t) = (\frac{1}{4}r, -r + s, -\frac{19}{2}r + 6s + t)$.

(d) Let $\mathscr{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be the standard basis for $\mathbb{R}^3$. Find the coordinate representations

$$A = [T]_{\mathscr{B},\mathscr{B}} \in \mathbb{R}^{3\times 3} \quad \text{and} \quad B = [T^{-1}]_{\mathscr{B},\mathscr{B}} \in \mathbb{R}^{3\times 3}$$

of T and T^{-1} with respect to the basis $\mathscr{B}$ on both the domain and the codomain.
SOLUTION. Since $T(1, 0, 0) = (2, 4, 2)$, $T(0, 1, 0) = (0, -1, 3)$, $T(0, 0, 1) = (0, 0, -1)$, we get

$$A = [T]_{\mathscr{B},\mathscr{B}} = \begin{bmatrix} 2 & 0 & 0 \\ 4 & -1 & 0 \\ 2 & 3 & -1 \end{bmatrix}.$$

Likewise, since $T^{-1}(1, 0, 0) = (\frac{1}{2}, 2, 7)$, $T^{-1}(0, 1, 0) = (0, -1, -3)$, $T(0, 0, 1) = (0, 0, -1)$, we get

$$B = [T^{-1}]_{\mathscr{B},\mathscr{B}} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 2 & -1 & 0 \\ 7 & -3 & -1 \end{bmatrix}.$$

(e) Are the matrices A and B nonsingular (a.k.a. invertible)? If so, what are A^{-1} and B^{-1}?
SOLUTION. Note that $AB = I = BA$. So A and B are both nonsingular and $A^{-1} = B$, $B^{-1} = A$.

小红书
小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 8 SOLUTIONS

1. Let $A \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

(a) Let $\mathbf{x}_0 \in \mathbb{R}^n$ be a solution of $A\mathbf{x} = \mathbf{b}$. Show that every solution of $A\mathbf{x} = \mathbf{b}$ is of the form

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{z}$$

where $\mathbf{z} \in \text{nullsp}(A)$.

SOLUTION. Since $\mathbf{x}_0 \in \mathbb{R}^n$ is a solution of $A\mathbf{x} = \mathbf{b}$, we have

$$A\mathbf{x}_0 = \mathbf{b}. \tag{1.1}$$

Let $\mathbf{x}$ be any solution of $A\mathbf{x} = \mathbf{b}$. Then

$$A\mathbf{x} = \mathbf{b}. \tag{1.2}$$

Let $\mathbf{z} = \mathbf{x} - \mathbf{x}_0$. By (1.1) and (1.2), we see htat

$$A\mathbf{z} = A(\mathbf{x} - \mathbf{x}_0) = A\mathbf{x} - A\mathbf{x}_0 == \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

Hence $\mathbf{z} \in \text{nullsp}(A)$, as required.

(b) Suppose $\mathbf{b} \neq \mathbf{0}$. Let $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k \in \mathbb{R}^n$ be solutions of $A\mathbf{x} = \mathbf{b}$, ie. $A\mathbf{x}_i = \mathbf{b}$ for all $i \in \{1, \ldots, k\}$. Show that the linear combination

$$\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + \cdots + \lambda_k\mathbf{x}_k$$

is also a solution to $A\mathbf{x} = \mathbf{b}$ if and only if

$$\lambda_1 + \lambda_2 + \cdots + \lambda_k = 1.$$

SOLUTION. By the given assumption,

$$A\mathbf{x}_1 = \mathbf{b}, \ldots, A\mathbf{x}_k = \mathbf{b}.$$

Now if $\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + \cdots + \lambda_k\mathbf{x}_k$ is also a solution, then

$$A(\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + \cdots + \lambda_k\mathbf{x}_k) = \mathbf{b}.$$

Therefore

$$\lambda_1 A\mathbf{x}_1 + \lambda_2 A\mathbf{x}_2 + \cdots + \lambda_k A\mathbf{x}_k = \mathbf{b}.$$

Therefore

$$\lambda_1\mathbf{b} + \lambda_2\mathbf{b} + \cdots + \lambda_k\mathbf{b} = \mathbf{b}.$$

Therefore

$$(\lambda_1 + \lambda_2 + \cdots + \lambda_k - 1)\mathbf{b} = \mathbf{0}.$$

Since $\mathbf{b} \neq \mathbf{0}$, we must have

$$\lambda_1 + \lambda_2 + \cdots + \lambda_k = 1.$$

Conversely, suppose $\lambda_1 + \lambda_2 + \cdots + \lambda_k = 1$, then

$$\begin{aligned} A(\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + \cdots + \lambda_k\mathbf{x}_k) &= \lambda_1 A\mathbf{x}_1 + \lambda_2 A\mathbf{x}_2 + \cdots + \lambda_k A\mathbf{x}_k \\ &= \lambda_1\mathbf{b} + \lambda_2\mathbf{b} + \cdots + \lambda_k\mathbf{b} \\ &= (\lambda_1 + \lambda_2 + \cdots + \lambda_k)\mathbf{b} \\ &= \mathbf{b}. \end{aligned}$$

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小红书
小红书号:3189205503 qq

(c) Show that

$$\operatorname{nullsp}(A^\top A) = \operatorname{nullsp}(A).$$

SOLUTION. Let $\mathbf{x} \in \operatorname{nullsp}(A)$. Then $A\mathbf{x} = \mathbf{0}$. Therefore, $A^\top A\mathbf{x} = A^\top \mathbf{0} = \mathbf{0}$. Therefore $\mathbf{x} \in \operatorname{nullsp}(A^\top A)$. So $\operatorname{nullsp}(A) \subseteq \operatorname{nullsp}(A^\top A)$. Conversely, let $\mathbf{x} \in \operatorname{nullsp}(A^\top A)$. Then $A^\top A\mathbf{x} = \mathbf{0}$. Multiplying on the left by $\mathbf{x}^\top$, we get

$$\mathbf{x}^\top A^\top A\mathbf{x} = \mathbf{x}^\top \mathbf{0} = 0.$$

Observe that the LHS may be written as

$$\mathbf{x}^\top A^\top A\mathbf{x} = (A\mathbf{x})^\top A\mathbf{x} = \langle A\mathbf{x}, A\mathbf{x}\rangle = \|A\mathbf{x}\|_2^2$$

and the last term is 0 iff $A\mathbf{x} = \mathbf{0}$ by a property of norms. Therefore $\mathbf{x} \in \operatorname{nullsp}(A)$. So $\operatorname{nullsp}(A^\top A) \subseteq \operatorname{nullsp}(A)$.

(d) Show that

$$A^\top A\mathbf{x} = A^\top \mathbf{b}$$

always have a solution (even if $A\mathbf{x} = \mathbf{b}$ has no solution).

SOLUTION. By a theorem in the lecture (relating consistency of linear systems to column space), $A^\top A\mathbf{x} = A^\top \mathbf{b}$ has a solution iff $A^\top \mathbf{b} \in \operatorname{colsp}(A^\top A)$. Applying the rank-nullity theorem to $A^\top A$, we get

$$\operatorname{rank}(A^\top A) = n - \operatorname{nullity}(A^\top A). \tag{1.3}$$

By part (c),

$$\operatorname{nullity}(A^\top A) = \operatorname{nullity}(A). \tag{1.4}$$

Applying the rank-nullity theorem to A yields

$$\operatorname{nullity}(A) = n - \operatorname{rank}(A). \tag{1.5}$$

By the equality of row rank and column rank, we have

$$\operatorname{rank}(A) = \operatorname{rank}(A^\top). \tag{1.6}$$

Combining (1.3)–(1.6), we get

$$\dim(\operatorname{colsp}(A^\top A)) = \dim(\operatorname{colsp}(A^\top)). \tag{1.7}$$

Now observe that if $\mathbf{y} \in \operatorname{colsp}(A^\top A)$, then $\mathbf{y} = A^\top A\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n$ and so $\mathbf{y} = A^\top \mathbf{w}$ for $\mathbf{w} = A\mathbf{x} \in \mathbb{R}^m$. In other words,

$$\operatorname{colsp}(A^\top A) \subseteq \operatorname{colsp}(A^\top). \tag{1.8}$$

(1.7) and (1.8) together implies that

$$\operatorname{colsp}(A^\top A) = \operatorname{colsp}(A^\top).$$

In particular, $A^\top \mathbf{b} \in \operatorname{colsp}(A^\top)$ and so $A^\top \mathbf{b} \in \operatorname{colsp}(A^\top A)$.

(e) Show that if $A\mathbf{x} = \mathbf{0}$ has a non-zero complex solution, ie. there exists $\mathbf{z} \in \mathbb{C}^m$, $\mathbf{z} \neq \mathbf{0}$, such that $A\mathbf{z} = \mathbf{0}$, then there exists a non-zero real solution.

SOLUTION. Suppose

$$\mathbf{z} = \begin{bmatrix} x_1 + iy_1 \\ \vdots \\ x_m + iy_m \end{bmatrix} \in \mathbb{C}^m.$$

Then we may write $\mathbf{z} = \mathbf{x} + i\mathbf{y}$ where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \in \mathbb{R}^m \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \in \mathbb{R}^m.$$

小红书
小红书号:3189205503 qq

So $A\mathbf{x} + iA\mathbf{y} = A(\mathbf{x} + i\mathbf{y}) = A\mathbf{z} = \mathbf{0}$ implies that

$$A\mathbf{x} = \mathbf{0} \quad \text{and} \quad A\mathbf{y} = \mathbf{0}.$$

Since $\mathbf{z} \neq \mathbf{0}$, either $\mathbf{x} \neq \mathbf{0}$ or $\mathbf{y} \neq \mathbf{0}$. So either $\mathbf{x}$ or $\mathbf{y}$ (or both) must be a non-zero real solution.

2. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbb{R}$. Let $\|\cdot\|$ be the norm induced by the inner product. Let $\mathbf{u}, \mathbf{v} \in V$. Show that $\|\mathbf{u}+\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$ if and only if $\|\alpha\mathbf{u}+\beta\mathbf{v}\| = \alpha\|\mathbf{u}\| + \beta\|\mathbf{v}\|$ for all $\alpha, \beta \geq 0$.

SOLUTION. Suppose $\|\mathbf{u}+\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$. Let $\alpha, \beta \geq 0$. By the triangle inequality and scaling property of a norm,

$$\|\alpha\mathbf{u} + \beta\mathbf{v}\| \leq \|\alpha\mathbf{u}\| + \|\beta\mathbf{v}\| = |\alpha|\|\mathbf{u}\| + |\beta|\|\mathbf{v}\| = \alpha\|\mathbf{u}\| + \beta\|\mathbf{v}\|.$$

On the other hand,

$$\begin{aligned}
\|\alpha\mathbf{u} + \beta\mathbf{v}\| &= \|\beta(\mathbf{u} + \mathbf{v}) - (\beta - \alpha)\mathbf{u}\| \\
&\geq \beta\|\mathbf{u} + \mathbf{v}\| - (\beta - \alpha)\|\mathbf{u}\| \\
&= \beta\|\mathbf{u}\| + \beta\|\mathbf{v}\| - (\beta - \alpha)\|\mathbf{u}\| \\
&= \alpha\|\mathbf{u}\| + \beta\|\mathbf{v}\|.
\end{aligned}$$

Hence

$$\|\alpha\mathbf{u} + \beta\mathbf{v}\| = \alpha\|\mathbf{u}\| + \beta\|\mathbf{v}\|.$$

Conversely, suppose $\|\alpha\mathbf{u} + \beta\mathbf{v}\| = \alpha\|\mathbf{u}\| + \beta\|\mathbf{v}\|$ for all $\alpha, \beta \geq 0$. Then setting $\alpha = \beta = 1$ yields

$$\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|.$$

3. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbb{R}$. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n \in V$ and let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ be the matrix with

$$a_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$$

for $i, j = 1, \ldots, n$. Show that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent if and only if $\text{nullity}(A) = 0$.

SOLUTION. Suppose there exist $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{R}$ such that

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n = \mathbf{0}.$$

Then taking inner product with each $\mathbf{v}_j$ yields

$$\langle \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n, \mathbf{v}_j \rangle = \langle \mathbf{0}, \mathbf{v}_j \rangle,$$

that is,

$$\langle \mathbf{v}_1, \mathbf{v}_j \rangle\lambda_1 + \langle \mathbf{v}_2, \mathbf{v}_j \rangle\lambda_2 + \cdots + \langle \mathbf{v}_n, \mathbf{v}_j \rangle\lambda_n = 0.$$

Doing this for $j = 1, \ldots, n$, we get

$$\begin{aligned}
\langle \mathbf{v}_1, \mathbf{v}_1 \rangle\lambda_1 + \langle \mathbf{v}_2, \mathbf{v}_1 \rangle\lambda_2 + \cdots + \langle \mathbf{v}_n, \mathbf{v}_1 \rangle\lambda_n &= 0, \\
\langle \mathbf{v}_1, \mathbf{v}_2 \rangle\lambda_1 + \langle \mathbf{v}_2, \mathbf{v}_2 \rangle\lambda_2 + \cdots + \langle \mathbf{v}_n, \mathbf{v}_2 \rangle\lambda_n &= 0, \\
&\cdots \\
\langle \mathbf{v}_1, \mathbf{v}_n \rangle\lambda_1 + \langle \mathbf{v}_2, \mathbf{v}_n \rangle\lambda_2 + \cdots + \langle \mathbf{v}_n, \mathbf{v}_n \rangle\lambda_n &= 0.
\end{aligned}$$

Note that this is a system of linear equations with $\lambda_1, \lambda_2, \ldots, \lambda_n$ as the unknowns. Writting in matrix form, we get

$$\begin{bmatrix}
\langle \mathbf{v}_1, \mathbf{v}_1 \rangle & \langle \mathbf{v}_1, \mathbf{v}_2 \rangle & \cdots & \langle \mathbf{v}_1, \mathbf{v}_n \rangle \\
\langle \mathbf{v}_2, \mathbf{v}_1 \rangle & \langle \mathbf{v}_2, \mathbf{v}_2 \rangle & \cdots & \langle \mathbf{v}_2, \mathbf{v}_n \rangle \\
\vdots & \vdots & \ddots & \vdots \\
\langle \mathbf{v}_n, \mathbf{v}_1 \rangle & \langle \mathbf{v}_n, \mathbf{v}_2 \rangle & \cdots & \langle \mathbf{v}_n, \mathbf{v}_n \rangle
\end{bmatrix}
\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{bmatrix} =
\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

that is,

$$A\boldsymbol{\lambda} = \mathbf{0},$$

小红书
小红书号:3189205503 qq

or equivalently,

$$\boldsymbol{\lambda} \in \text{nullsp}(A).$$

In other words, we have shown that every $\boldsymbol{\lambda} = [\lambda_1, \lambda_2, \ldots, \lambda_n]^\top$ that gives a zero linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is an element of $\text{nullsp}(A)$. Hence $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent if and only if $\text{nullity}(A) = 0$.

4. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbb{R}$. Show that if $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \in V$ are pairwise orthogonal unit vectors, ie. $\|\mathbf{u}_i\| = 1$ for all $i = 1, \ldots, n$, and $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $i \neq j$, then

$$\sum_{i=1}^{n} |\langle \mathbf{v}, \mathbf{u}_i \rangle|^2 \leq \|\mathbf{v}\|^2$$

for all $\mathbf{v} \in V$. [Note: this is called the Bessel's inequality; when $n = 1$, this reduces to the Cauchy-Schwartz inequality].

SOLUTION. We consider the vector

$$\mathbf{w} = \mathbf{v} - \sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i.$$

From the trivial inequality $\|\mathbf{w}\|^2 \geq 0$, we get

$$\begin{aligned}
0 &\leq \left\| \mathbf{v} - \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i \right\|^2 \\
&= \left\langle \mathbf{v} - \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i, \mathbf{v} - \textstyle\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{u}_j \rangle \mathbf{u}_j \right\rangle \\
&= \left\langle \mathbf{v}, \mathbf{v} - \textstyle\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{u}_j \rangle \mathbf{u}_j \right\rangle - \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \left\langle \mathbf{u}_i, \mathbf{v} - \textstyle\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{u}_j \rangle \mathbf{u}_j \right\rangle \\
&= \langle \mathbf{v}, \mathbf{v} \rangle - \textstyle\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{u}_j \rangle \langle \mathbf{v}, \mathbf{u}_j \rangle - \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \left[\langle \mathbf{u}_i, \mathbf{v} \rangle - \textstyle\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{u}_j \rangle \langle \mathbf{u}_i, \mathbf{u}_j \rangle \right] \\
&= \|\mathbf{v}\|^2 - \textstyle\sum_{j=1}^{n} |\langle \mathbf{v}, \mathbf{u}_j \rangle|^2 - \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \langle \mathbf{u}_i, \mathbf{v} \rangle + \textstyle\sum_{i=1}^{n} \textstyle\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \langle \mathbf{v}, \mathbf{u}_j \rangle \langle \mathbf{u}_i, \mathbf{u}_j \rangle
\end{aligned}$$

Now observe that the term $\langle \mathbf{u}_i, \mathbf{u}_j \rangle$ in the double sum is 1 if $i = j$ and 0 otherwise. Hence,

$$0 \leq \|\mathbf{v}\|^2 - \textstyle\sum_{j=1}^{n} |\langle \mathbf{v}, \mathbf{u}_j \rangle|^2 - \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \langle \mathbf{u}_i, \mathbf{v} \rangle + \textstyle\sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \langle \mathbf{v}, \mathbf{u}_i \rangle.$$

The last two sums cancel each other since $\langle \mathbf{u}_i, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u}_i \rangle$. This leaves us with

$$0 \leq \|\mathbf{v}\|^2 - \textstyle\sum_{j=1}^{n} |\langle \mathbf{v}, \mathbf{u}_j \rangle|^2$$

or

$$\textstyle\sum_{i=1}^{n} |\langle \mathbf{v}, \mathbf{u}_i \rangle|^2 \leq \|\mathbf{v}\|^2.$$

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小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 9 SOLUTIONS

In the following V will denote a finite-dimensional vector space over $\mathbb{R}$ that is also an inner product space with inner product denoted by $\langle \cdot, \cdot \rangle$. The norm induced by $\langle \cdot, \cdot \rangle$ will be denoted $\| \cdot \|$.

1. Let S be a subset (not necessarily a subspace) of V. We define the *orthogonal annihilator* of S, denoted $S^\perp$, to be the set

$$S^\perp = \{\mathbf{v} \in V \mid \langle \mathbf{v}, \mathbf{w} \rangle = 0 \text{ for all } \mathbf{w} \in S\}.$$

(a) Show that $S^\perp$ is always a subspace of V.
SOLUTION. Let $\mathbf{v}_1, \mathbf{v}_2 \in S^\perp$. Then $\langle \mathbf{v}_1, \mathbf{w} \rangle = 0$ and $\langle \mathbf{v}_2, \mathbf{w} \rangle = 0$ for all $\mathbf{w} \in S$. So for any $\alpha, \beta \in \mathbb{R}$,

$$\langle \alpha\mathbf{v}_1 + \beta\mathbf{v}_2, \mathbf{w} \rangle = \alpha\langle \mathbf{v}_1, \mathbf{w} \rangle + \beta\langle \mathbf{v}_2, \mathbf{w} \rangle = 0$$

for all $\mathbf{w} \in S$. Hence $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 \in S^\perp$.

(b) Show that $S \subseteq (S^\perp)^\perp$.
SOLUTION. Let $\mathbf{w} \in S$. For any $\mathbf{v} \in S^\perp$, we have $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ by definition of $S^\perp$. Since this is true for all $\mathbf{v} \in S^\perp$ and $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle$, we see that $\langle \mathbf{w}, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in S^\perp$, ie. $\mathbf{w} \in S^\perp$.

(c) Show that $\operatorname{span}(S) \subseteq (S^\perp)^\perp$.
SOLUTION. Since $(S^\perp)^\perp$ is a subspace by (a) (it is the orthogonal annihilator of $S^\perp$) and $S \subseteq (S^\perp)^\perp$ by (b), we have $\operatorname{span}(S) \subseteq (S^\perp)^\perp$.

(d) Show that if S_1 and S_2 are subsets of V and $S_1 \subseteq S_2$, then $S_2^\perp \subseteq S_1^\perp$.
SOLUTION. Let $\mathbf{v} \in S_2^\perp$. Then $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{w} \in S_2$ and so for all $\mathbf{w} \in S_1$ (since $S_1 \subseteq S_2$). Hence $\mathbf{v} \in S_1^\perp$.

(e) Show that $((S^\perp)^\perp)^\perp = S^\perp$.
SOLUTION. Applying (d) to (b) with $S_1 = S$ and $S_2 = (S^\perp)^\perp$, we get

$$((S^\perp)^\perp)^\perp \subseteq S^\perp.$$

Apply (c) to $S^\perp$, we get

$$\operatorname{span}(S^\perp) \subseteq ((S^\perp)^\perp)^\perp$$

but by (a), $\operatorname{span}(S^\perp) = S^\perp$. Hence we get equality.

(f) Show that either $S \cap S^\perp$ must be either the empty set $\varnothing$ or the zero subspace $\{\mathbf{0}_V\}$.
SOLUTION. If $S \cap S^\perp \neq \varnothing$, then let $\mathbf{v} \in S \cap S^\perp$. Since $\mathbf{v} \in S^\perp$, we have $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{w} \in S$. Since $\mathbf{v} \in S$, in particular, $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 0$, implying that $\mathbf{v} = \mathbf{0}_V$. In other words, the only vector in $S \cap S^\perp$ is $\mathbf{0}_V$. So $S \cap S^\perp = \{\mathbf{0}_V\}$. On the other hand, if $\mathbf{0}_V \notin S$, then $\mathbf{0}_V \notin S \cap S^\perp$ and so $S \cap S^\perp = \varnothing$ (if not, then the previous argument gives a contradiction).

2. Let W be a subspace of V. The subspace $W^\perp$ is called the *orthogonal complement* of W.

(a) Show that $V = W \oplus W^\perp$.
SOLUTION. Since W and $W^\perp$ are both subspaces (the latter by Problem **1**(a)), $\mathbf{0}_V \in W$ and $\mathbf{0}_V \in W^\perp$. So by Problem **1**(f), we have that $W \cap W^\perp = \{\mathbf{0}_V\}$. It remains to show that

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小红书
小红书号:3189205503 qq

$W + W^\perp = V$. Clearly $W + W^\perp \subseteq V$. For the converse, let $\mathbf{v} \in V$ and let $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$ be an orthonormal basis of W. Consider

$$\mathbf{x} := \langle \mathbf{v}, \mathbf{w}_1 \rangle \mathbf{w}_1 + \langle \mathbf{v}, \mathbf{w}_2 \rangle \mathbf{w}_2 + \cdots + \langle \mathbf{v}, \mathbf{w}_r \rangle \mathbf{w}_r$$

and

$$\mathbf{y} := \mathbf{v} - \mathbf{x}.$$

Clearly $\mathbf{x} \in W$. We claim that $\mathbf{y} \in W^\perp$. For any $\mathbf{w} \in W$, we could write

$$\mathbf{w} = \langle \mathbf{w}, \mathbf{w}_1 \rangle \mathbf{w}_1 + \langle \mathbf{w}, \mathbf{w}_2 \rangle \mathbf{w}_2 + \cdots + \langle \mathbf{w}, \mathbf{w}_r \rangle \mathbf{w}_r$$

since $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$ is an orthonormal basis of W. So

$$\begin{aligned}
\langle \mathbf{y}, \mathbf{w} \rangle &= \langle \mathbf{v} - \mathbf{x}, \mathbf{w} \rangle \\
&= \langle \mathbf{v}, \mathbf{w} \rangle - \langle \mathbf{x}, \mathbf{w} \rangle \\
&= \Big\langle \mathbf{v}, \sum\nolimits_{i=1}^r \langle \mathbf{w}, \mathbf{w}_i \rangle \mathbf{w}_i \Big\rangle - \Big\langle \sum\nolimits_{i=1}^r \langle \mathbf{v}, \mathbf{w}_i \rangle \mathbf{w}_i, \mathbf{w} \Big\rangle \\
&= \sum\nolimits_{i=1}^r \langle \mathbf{v}, \mathbf{w}_i \rangle \langle \mathbf{w}, \mathbf{w}_i \rangle - \sum\nolimits_{i=1}^r \langle \mathbf{v}, \mathbf{w}_i \rangle \langle \mathbf{w}_i, \mathbf{w} \rangle \\
&= 0
\end{aligned}$$

since $\langle \mathbf{w}, \mathbf{w}_i \rangle = \langle \mathbf{w}_i, \mathbf{w} \rangle$. Hence $\mathbf{v} = \mathbf{x} + \mathbf{y} \in W + W^\perp$.

(b) Show that $W = (W^\perp)^\perp$.

Solution. By Problem **1**(b), we have $W \subseteq (W^\perp)^\perp$. But by (a), we have

$$W \oplus W^\perp = V = W^\perp \oplus (W^\perp)^\perp$$

and so

$$\dim(W) + \dim(W^\perp) = \dim(W^\perp) + \dim((W^\perp)^\perp)$$

and so

$$\dim(W) = \dim((W^\perp)^\perp).$$

Hence $W = (W^\perp)^\perp$.

3. Let $A \in \mathbb{R}^{m \times n}$ and regard $A : \mathbb{R}^n \to \mathbb{R}^m$ as a linear transformation in the usual way. Show that

$$\operatorname{nullsp}(A^\top) = (\operatorname{colsp}(A))^\perp$$

and

$$\operatorname{rowsp}(A) = (\operatorname{nullsp}(A))^\perp.$$

Hence deduce that

$$\mathbb{R}^m = \operatorname{nullsp}(A^\top) \oplus \operatorname{colsp}(A)$$

and

$$\mathbb{R}^n = \operatorname{rowsp}(A) \oplus \operatorname{nullsp}(A).$$

Solution. Clearly $\operatorname{colsp}(A) = \{A\mathbf{x} \in \mathbb{R}^m \mid \mathbf{x} \in \mathbb{R}^n\}$. So if $\mathbf{v} \in (\operatorname{colsp}(A))^\perp$, then

$$\langle \mathbf{v}, A\mathbf{x} \rangle = 0$$

for all $\mathbf{x} \in \mathbb{R}^n$. So

$$\langle A^\top \mathbf{v}, \mathbf{x} \rangle = (A^\top \mathbf{v})^\top \mathbf{x} = \mathbf{v}^\top A\mathbf{x} = \langle \mathbf{v}, A\mathbf{x} \rangle = 0$$

for all $\mathbf{x} \in \mathbb{R}^n$. In particular, we may pick $\mathbf{x} = A^\top \mathbf{v}$ to get

$$\|A^\top \mathbf{v}\|^2 = \langle A^\top \mathbf{v}, A^\top \mathbf{v} \rangle = 0,$$

giving us

$$A^\top \mathbf{v} = \mathbf{0}.$$

Hence $\mathbf{v} \in \operatorname{nullsp}(A^\top)$. Conversely, if $\mathbf{v} \in \operatorname{nullsp}(A^\top)$, then

$$\langle \mathbf{v}, A\mathbf{x} \rangle = \mathbf{v}^\top A\mathbf{x} = (A^\top \mathbf{v})^\top \mathbf{x} = \mathbf{0}^\top \mathbf{x} = 0$$

小红书
小红书号:3189205503 qq

for all $\mathbf{x} \in \mathbb{R}^n$. So $\mathbf{v} \in (\text{colsp}(A))^\perp$. This proves the first equality. For the second equality, just note that

$$\text{rowsp}(A) = \text{colsp}(A^\top)$$

and apply the first equality to get

$$(\text{colsp}(A^\top))^\perp = \text{nullsp}(A),$$

and then apply Problem **2**(b) to get

$$\text{rowsp}(A) = ((\text{rowsp}(A))^\perp)^\perp = (\text{nullsp}(A))^\perp.$$

The next two inequalities are immediate consequences of Problem **2**(a).

4. Let $A, B \in \mathbb{R}^{n\times n}$. Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^n$.

(a) Show that if $\langle A\mathbf{x}, \mathbf{y}\rangle = 0$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then $A = O$, the zero matrix.
SOLUTION. Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. Observe that if $A = [a_{ij}]_{i,j=1}^n$, then

$$a_{ij} = \mathbf{e}_i^\top A\mathbf{e}_j = \langle \mathbf{e}_i, A\mathbf{e}_j\rangle = \langle A\mathbf{e}_j, \mathbf{e}_i\rangle = 0$$

by letting $\mathbf{x}$ and $\mathbf{y}$ be $\mathbf{e}_i$ and $\mathbf{e}_j$ respectively. Since this is true for all i and j, we have that $A = O$.

(b) Is it true that if $\langle A\mathbf{x}, \mathbf{x}\rangle = 0$ for all $\mathbf{x} \in \mathbb{R}^n$, then $A = O$, the zero matrix?
SOLUTION. Recall notation from Homework **4**, Problem **1**(d). Let $A \in \wedge^2(\mathbb{R}^n)$, ie. $A^\top = -A$. Then

$$\langle A\mathbf{x}, \mathbf{x}\rangle = \mathbf{x}^\top A^\top \mathbf{x} = -\mathbf{x}^\top A\mathbf{x} = -\langle \mathbf{x}, A\mathbf{x}\rangle = -\langle A\mathbf{x}, \mathbf{x}\rangle,$$

implying that $\langle A\mathbf{x}, \mathbf{x}\rangle = 0$. So any non-zero skew-symmetric matrix would be a counter example. An explicit 2×2 example is given by

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

(c) Is it true that if $\langle A\mathbf{x}, \mathbf{x}\rangle = \langle B\mathbf{x}, \mathbf{x}\rangle$ for all $\mathbf{x} \in \mathbb{R}^n$, then $A = B$?
SOLUTION. Any two distinct skew-symmetric matrices would provide a counter example. An explicit 2×2 example is given by

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

5. Let $\mathscr{B}_1 = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ and $\mathscr{B}_2 = \{\mathbf{y}_1, \ldots, \mathbf{y}_n\}$ be two bases of $\mathbb{R}^n$. $\mathscr{B}_1$ and $\mathscr{B}_2$ are not necessarily orthonormal bases.

(a) Show that there exists an orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ such that

$$Q\mathbf{x}_i = \mathbf{y}_i, \quad i = 1, \ldots, n,$$

if and only if

$$\langle \mathbf{x}_i, \mathbf{x}_j\rangle = \langle \mathbf{y}_i, \mathbf{y}_j\rangle, \quad i, j = 1, \ldots, n.$$

SOLUTION. By a theorem we have proved in the lectures, an orthogonal matrix Q preserves inner product and so satisfies

$$\langle \mathbf{y}_i, \mathbf{y}_j\rangle = \langle Q\mathbf{x}_i, Q\mathbf{x}_j\rangle = \langle \mathbf{x}_i, \mathbf{x}_j\rangle.$$

It remains to show that if the latter condition is satisfied, then such an orthogonal matrix may be defined. First, observe that since $\mathscr{B}_1$ and $\mathscr{B}_2$ are bases, the equations

$$Q\mathbf{x}_i = \mathbf{y}_i, \quad i = 1, \ldots, n,$$

小红书
小红书号:3189205503 qq

defines the matrix Q uniquely[1]. We only need to verify that Q is orthogonal. Let $\mathbf{x} \in \mathbb{R}^n$ and let

$$\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{x}_i$$

be the representation of $\mathbf{x}$ in terms of $\mathscr{B}_1$. Then

$$\begin{aligned}
\|Q\mathbf{x}\|^2 &= \langle Q\mathbf{x}, Q\mathbf{x}\rangle \\
&= \left\langle \sum_{i=1}^n \alpha_i Q\mathbf{x}_i, \sum_{j=1}^n \alpha_j Q\mathbf{x}_j \right\rangle \\
&= \left\langle \sum_{i=1}^n \alpha_i \mathbf{y}_i, \sum_{j=1}^n \alpha_j \mathbf{y}_j \right\rangle \\
&= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \langle \mathbf{y}_i, \mathbf{y}_j \rangle \\
&= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle \\
&= \left\langle \sum_{i=1}^n \alpha_i \mathbf{x}_i, \sum_{j=1}^n \alpha_j \mathbf{x}_j \right\rangle \\
&= \langle \mathbf{x}, \mathbf{x}\rangle \\
&= \|\mathbf{x}\|^2.
\end{aligned}$$

Since this is true for all $\mathbf{x} \in \mathbb{R}^n$, ie. Q preserves norm, the aforementioned theorem in the lectures imply that Q is orthogonal.

(b) Show that if $\mathscr{B}_1$ is an orthonormal basis and if $Q \in \mathbb{R}^{n\times n}$ is an orthogonal matrix, then

$$\{Q\mathbf{x}_1, \ldots, Q\mathbf{x}_n\}$$

is also an orthonormal basis.

Solution. Clearly $Q\mathbf{x}_1, \ldots, Q\mathbf{x}_n$ are linearly independent since if

$$\alpha_1 Q\mathbf{x}_1 + \cdots + \alpha_n Q\mathbf{x}_n = \mathbf{0},$$

then

$$Q(\alpha_1 \mathbf{x}_1 + \cdots + \alpha_n \mathbf{x}_n) = \mathbf{0},$$

and so

$$\alpha_1 \mathbf{x}_1 + \cdots + \alpha_n \mathbf{x}_n = Q^\top \mathbf{0} = \mathbf{0}$$

and so $\alpha_1 = \cdots = \alpha_n = 0$ by the linear independence of $\mathbf{x}_1, \ldots, \mathbf{x}_n$. Hence $\{Q\mathbf{x}_1, \ldots, Q\mathbf{x}_n\}$ is a basis (since the set has size $n = \dim(\mathbb{R}^n)$). The orthonormality follows from the fact that Q preserves inner product and that $\mathbf{x}_1, \ldots, \mathbf{x}_n$ form an orthonormal basis:

$$\langle Q\mathbf{x}_i, Q\mathbf{x}_j\rangle = \langle \mathbf{x}_i, \mathbf{x}_j\rangle = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

[1] In fact, Q is just the matrix whose coordinate representation with respect to $\mathscr{B}_1$ and $\mathscr{B}_2$ is the identity matrix, ie. $[Q]_{\mathscr{B}_1,\mathscr{B}_2} = I$. Explicitly, if we let X and Y be the matrices whose columns are $\mathbf{x}_1, \ldots, \mathbf{x}_n$ and $\mathbf{y}_1, \ldots, \mathbf{y}_n$ respectively, then $Q = YX^{-1}$.

小红书
小红书号:3189205503 qq

MATH 110: LINEAR ALGEBRA
FALL 2007/08
PROBLEM SET 10 SOLUTIONS

Elements in $\mathbb{R}^n$ will always be written as column vectors. The inner product on $\mathbb{R}^n$ will be denoted by $\langle \cdot, \cdot \rangle$ and its induced norm will be denoted $\| \cdot \|_2$. I will denote the identity matrix. $A \in \mathbb{R}^{m \times n}$ is called *full-rank* if $\operatorname{rank}(A) = \min\{m, n\}$. $M \in \mathbb{R}^{n \times n}$ is called *nonsymmetric* if $M^\top \neq M$.

1. A square matrix $M \in \mathbb{R}^{n \times n}$ is called *positive semidefinite* if

$$\mathbf{x}^\top M \mathbf{x} \geq 0$$

for all $\mathbf{x} \in \mathbb{R}^n$. $M \in \mathbb{R}^{n \times n}$ is called *positive definite* if (i) M is positive semidefinite; and (ii) $\mathbf{x}^\top M \mathbf{x} = 0$ only if $\mathbf{x} = \mathbf{0}$.

(a) Show that every positive definite matrix is nonsingular (ie. invertible).
SOLUTION. Let $\mathbf{x} \in \operatorname{nullsp}(M)$. Then $M\mathbf{x} = \mathbf{0}$. Then $\mathbf{x}^\top M \mathbf{x} = \mathbf{x}^\top \mathbf{0} = 0$. So $\mathbf{x} = \mathbf{0}$ since M is positive definite. Hence $\operatorname{nullsp}(M) = \{\mathbf{0}\}$. Hence M is nonsingular by a theorem in the lectures.

(b) Show that if M is positive semidefinite and $\lambda \in \mathbb{R}$ is an eigenvalue of M, then $\lambda \geq 0$.
SOLUTION. Let $\mathbf{x} \in \mathbb{R}^n$ be the eigenvector of M corresponding to λ, ie. $M\mathbf{x} = \lambda \mathbf{x}$. So

$$\mathbf{x}^\top M \mathbf{x} = \lambda \mathbf{x}^\top \mathbf{x} = \lambda \|\mathbf{x}\|_2^2.$$

Since $\mathbf{x} \neq \mathbf{0}$ (eigenvectors are non-zero), so $\|\mathbf{x}\|_2 > 0$, and we get

$$\lambda = \frac{\mathbf{x}^\top M \mathbf{x}}{\|\mathbf{x}\|_2^2} \geq 0$$

since M is positive semidefinite.

(c) Show that if M is positive definite and $\lambda \in \mathbb{R}$ is an eigenvalue of M, then $\lambda > 0$.
SOLUTION. Identical argument as above except that we have

$$\lambda = \frac{\mathbf{x}^\top M \mathbf{x}}{\|\mathbf{x}\|_2^2} > 0$$

since M is positive definite.

(d) Is it possible for a nonsymmetric matrix M to be positive semidefinite?
SOLUTION. By Homework **9**, Problem **4**(b), any non-zero skew-symmetric matrix A would satisfy

$$\mathbf{x}^\top A \mathbf{x} = 0$$

and $A^\top = -A \neq A$. So A is nonsymmetric and yet it is positive semidefinite.

(e) Is it possible for a nonsymmetric matrix M to be positive definite?
SOLUTION. Yes. We will deduce it as follows. Observe that (see solution to Homework **4**, Problem **1**),

$$M = \frac{1}{2}(M + M^\top) + \frac{1}{2}(M - M^\top).$$

Let

$$S := \frac{1}{2}(M + M^\top) \quad \text{and} \quad A := \frac{1}{2}(M - M^\top).$$

Note that S is symmetric while A is skew symmetric. Since $\mathbf{x}^\top A \mathbf{x} = 0$, we have

$$\mathbf{x}^\top M \mathbf{x} = \mathbf{x}^\top (S + A)\mathbf{x} = \mathbf{x}^\top S \mathbf{x} + \mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top S \mathbf{x}$$

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小红书
小红书号:3189205503 qq

for all $\mathbf{x} \in \mathbb{R}^n$. Hence the value of $\mathbf{x}^\top M\mathbf{x}$ depends only on $\mathbf{x}^\top S\mathbf{x}$. So as long as we pick an S that is positive definite and a non-zero skew symmetric matrix A, then

$$M := S + A$$

will be a counter example. For an explicit 2×2 counter example, pick

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

to get

$$M = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Observe that

$$\mathbf{x}^\top M\mathbf{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x^2 + y^2 > 0$$

for all $\mathbf{x} \neq \mathbf{0}$.

(f) Let $A \in \mathbb{R}^{m\times n}$. Show that $A^\top A$ and $AA^\top$ are positive semidefinite matrices. Hence deduce that singular values are always nonnegative.
SOLUTION. This just follows from the observation that

$$\mathbf{x}^\top A^\top A\mathbf{x} = (A\mathbf{x})^\top (A\mathbf{x}) = \|A\mathbf{x}\|_2^2 \geq 0$$

for all $\mathbf{x} \in \mathbb{R}^n$ and that

$$\mathbf{y}^\top AA^\top \mathbf{y} = (A^\top \mathbf{y})^\top (A^\top \mathbf{y}) = \|A^\top \mathbf{y}\|_2^2 \geq 0$$

for all $\mathbf{y} \in \mathbb{R}^m$.

(g) Let $A \in \mathbb{R}^{m\times n}$ be full-rank. Show that either $A^\top A$ or $AA^\top$ is a positive definite matrix.
SOLUTION. If A is full-rank, then $\operatorname{rank}(A) = \min\{m, n\}$. If $m \geq n$, then $\operatorname{rank}(A) = n$. By the rank-nullity theorem,

$$\operatorname{nullity}(A) = n - \operatorname{rank}(A) = 0.$$

If $\mathbf{x}^\top A^\top A\mathbf{x} = 0$, then $\|A\mathbf{x}\|_2^2 = 0$; so $A\mathbf{x} = \mathbf{0}$; so $\mathbf{x} \in \operatorname{nullity}(A)$; so $\mathbf{x} = \mathbf{0}$. Hence $A^\top A$ is positive definite. On the other hand, if $m < n$, then $\operatorname{rank}(A) = m$; so $\operatorname{rank}(A^\top) = m$. By the rank-nullity theorem,

$$\operatorname{nullity}(A^\top) = m - \operatorname{rank}(A^\top) = 0.$$

If $\mathbf{y}^\top AA^\top \mathbf{y} = 0$, then $\|A^\top \mathbf{y}\|_2^2 = 0$; so $A^\top \mathbf{y} = \mathbf{0}$; so $\mathbf{y} \in \operatorname{nullity}(A^\top)$; so $\mathbf{y} = \mathbf{0}$. Hence $AA^\top$ is positive definite.

2. Let $\mathbf{u} \in \mathbb{R}^n$, $\mathbf{u} \neq \mathbf{0}$. Consider the *Householder* matrix $H_\mathbf{u} \in \mathbb{R}^{n\times n}$ defined by

$$H_\mathbf{u} = I - \frac{2\mathbf{u}\mathbf{u}^\top}{\|\mathbf{u}\|_2^2}.$$

(*Note*: $\mathbf{u}$ is a column vector, ie. an $n \times 1$ matrix, so $\mathbf{u}^\top$ is a row vector, ie. an $1 \times n$ matrix, and so $\mathbf{u}\mathbf{u}^\top$ is an $n \times n$ matrix).

(a) Show that $H_\mathbf{u}$ is both symmetric and orthogonal.
SOLUTION. Since $(\mathbf{u}\mathbf{u}^\top)^\top = (\mathbf{u}^\top)^\top \mathbf{u}^\top = \mathbf{u}\mathbf{u}^\top$,

$$H_\mathbf{u}^\top = \left(I - \frac{2\mathbf{u}\mathbf{u}^\top}{\|\mathbf{u}\|_2^2}\right)^\top = I - \frac{2\mathbf{u}\mathbf{u}^\top}{\|\mathbf{u}\|_2^2} = H_\mathbf{u}.$$

小红书
小红书号:3189205503 qq

Hence

$$\begin{aligned} H_{\mathbf{u}}^{\top} H_{\mathbf{u}} &= H_{\mathbf{u}} H_{\mathbf{u}} \\ &= \left(I - \frac{2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2}\right)\left(I - \frac{2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2}\right) \\ &= I - \frac{2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2} - \frac{2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2} + \frac{4\mathbf{u}(\mathbf{u}^{\top}\mathbf{u})\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^4} \\ &= I - \frac{4\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2} + \frac{4\|\mathbf{u}\|_2^2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^4} \\ &= I. \end{aligned}$$

(b) Show that for any $\alpha \in \mathbb{R}$, $\alpha \neq 0$,

$$H_{\alpha\mathbf{u}} = H_{\mathbf{u}}.$$

In other words, $H_{\mathbf{u}}$ only depends on the 'direction' of $\mathbf{u}$ and not on its 'magnitude'.

SOLUTION. This follows from

$$H_{\alpha\mathbf{u}} = I - \frac{2\alpha\mathbf{u}(\alpha\mathbf{u})^{\top}}{\|\alpha\mathbf{u}\|_2^2} = I - \frac{2\alpha^2\mathbf{u}\mathbf{u}^{\top}}{\alpha^2\|\mathbf{u}\|_2^2} = I - \frac{2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2} = H_{\mathbf{u}}$$

for any $\alpha \in \mathbb{R}$, $\alpha \neq 0$.

(c) In general, given a matrix $M \in \mathbb{R}^{n\times n}$ and a vector $\mathbf{x} \in \mathbb{R}^n$, computing the matrix-vector product $M\mathbf{x}$ requires n inner products — one for each row of M with $\mathbf{x}$. Show that $H_{\mathbf{u}}\mathbf{x}$ can be computed using only two inner products.

SOLUTION. Let $\mathbf{x} \in \mathbb{R}^n$. Then

$$H_{\mathbf{u}}\mathbf{x} = \left(I - \frac{2\mathbf{u}\mathbf{u}^{\top}}{\|\mathbf{u}\|_2^2}\right)\mathbf{x} = \mathbf{x} - \left(\frac{2\mathbf{u}^{\top}\mathbf{x}}{\|\mathbf{u}\|_2^2}\right)\mathbf{u}.$$

Note that the coefficient of $\mathbf{u}$ involves only two inner products — $\mathbf{u}^{\top}\mathbf{x}$ and $\|\mathbf{u}\|_2^2 = \mathbf{u}^{\top}\mathbf{u}$.

(d) Given $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ where $\mathbf{a} \neq \mathbf{b}$ and $\|\mathbf{a}\|_2 = \|\mathbf{b}\|_2$. Find $\mathbf{u} \in \mathbb{R}^n$, $\mathbf{u} \neq \mathbf{0}$ such that

$$H_{\mathbf{u}}\mathbf{a} = \mathbf{b}.$$

SOLUTION. Let $\mathbf{u} = \mathbf{a} - \mathbf{b}$. Then $\mathbf{u} \neq \mathbf{0}$. By (b), we may assume that $\|\mathbf{u}\|_2 = \|\mathbf{a}-\mathbf{b}\|_2 = 1$, ie.

$$1 = \|\mathbf{a} - \mathbf{b}\|_2^2 = \|\mathbf{a}\|_2^2 - 2\mathbf{a}^{\top}\mathbf{b} + \|\mathbf{b}\|_2^2 = 2\|\mathbf{a}\|_2^2 - 2\mathbf{a}^{\top}\mathbf{b}$$

since $\|\mathbf{a}\|_2 = \|\mathbf{b}\|_2$. Then

$$\begin{aligned} H_{\mathbf{u}}\mathbf{a} &= \mathbf{a} - 2(\mathbf{a} - \mathbf{b})^{\top}\mathbf{a}(\mathbf{a} - \mathbf{b}) \\ &= (1 - 2\|\mathbf{a}\|_2^2 + 2\mathbf{b}^{\top}\mathbf{a})\mathbf{a} + (2\|\mathbf{a}\|_2^2 - 2\mathbf{b}^{\top}\mathbf{a})\mathbf{b} \\ &= (1 - 1)\mathbf{a} + \mathbf{b} \\ &= \mathbf{b}. \end{aligned}$$

(e) Show that $\mathbf{u}$ is an eigenvector of $H_{\mathbf{u}}$. What is the corresponding eigenvalue?

SOLUTION. First note that $\mathbf{u} \neq \mathbf{0}$. Since

$$H_{\mathbf{u}}\mathbf{u} = \mathbf{u} - \left(\frac{2\mathbf{u}^{\top}\mathbf{u}}{\|\mathbf{u}\|_2^2}\right)\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u},$$

$\mathbf{u}$ is an eigenvector of $H_{\mathbf{u}}$ corresponding to the eigenvalue -1.

(f) Recall the definition of orthogonal complement in Problem Set **9**. Let $U = \operatorname{span}\{\mathbf{u}\}$. Show that every $\mathbf{v} \in U^{\perp}$ is an eigenvector of $H_{\mathbf{u}}$. What are the corresponding eigenvalues? What is $\dim(U^{\perp})$?

小红书
小红书号:3189205503 qq

SOLUTION. Note that $\mathbf{v} \in U^\perp$ iff $\mathbf{v}^\top \mathbf{u} = 0$. Since

$$H_{\mathbf{u}}\mathbf{v} = \mathbf{v} - \left(\frac{2\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2^2}\right)\mathbf{u} = \mathbf{v},$$

every non-zero $\mathbf{v} \in U^\perp$ is a eigenvector of $H_{\mathbf{u}}$ corresponding to the eigenvalue 1. By Homework **9**, Problem **3**,

$$\dim(U^\perp) = \dim(\mathbb{R}^n) - \dim(U) = n - 1$$

since $\mathbb{R}^n = U \oplus U^\perp$.

(g) Using (d) and (e), find an orthogonal diagonalization of $H_{\mathbf{u}}$, ie. find an orthogonal matrix Q and a diagonal matrix Λ such that

$$H_{\mathbf{u}} = Q\Lambda Q^\top.$$

(*Hint*: you will need to use the Gram-Schmidt algorithm).

SOLUTION. By (e), $\mathbf{u}$ is an eigenvector of $H_{\mathbf{u}}$ corresponding to the eigenvalue -1; we let $\mathbf{u}_0 = \mathbf{u}/\|\mathbf{u}\|_2$. By (f), every non-zero vector in $U^\perp$ is an eigenvector of $H_{\mathbf{u}}$ corresponding to the eigenvalue 1. Since $\dim(U^\perp) = n - 1$, let $\{\mathbf{v}_1, \ldots, \mathbf{v}_{n-1}\}$ be a basis for $U^\perp$. Since $\mathbf{v}_1, \ldots, \mathbf{v}_{n-1}$ are linearly independent, we may apply the Gram-Schmidt algorithm to obtain an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_{n-1}$ of $U^\perp$, ie.

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j, \end{cases}$$

for $i, j \in \{1, \ldots, n-1\}$. However, since every vector in $U^\perp$ is orthogonal to $\mathbf{u}$, so

$$\langle \mathbf{u}_i, \mathbf{u}_0 \rangle = \langle \mathbf{u}_i, \mathbf{u} \rangle / \|\mathbf{u}\|_2 = 0$$

for all $i \in \{1, \ldots, n-1\}$. Hence $\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{n-1}$ form an orthonormal basis of $\mathbb{R}^n$ and so the matrix

$$Q := [\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{n-1}] \in \mathbb{R}^{n \times n}$$

is an orthogonal matrix. Since each $\mathbf{u}_i$ is also an eigenvector of $H_{\mathbf{u}}$, Q diagonalizes $H_{\mathbf{u}}$. In fact, we may easily see that

$$H_{\mathbf{u}} = Q\Lambda Q^\top$$

where

$$\Lambda = \operatorname{diag}(-1, 1, \ldots, 1).$$

3. (a) Let $Q \in \mathbb{R}^{n \times n}$ be an orthogonal matrix. Suppose $\lambda \in \mathbb{R}$ is an eigenvalue of Q. Show that either $\lambda = 1$ or $\lambda = -1$.

SOLUTION. Let $\mathbf{x}$ be the eigenvector of Q corresponding to λ. So $Q\mathbf{x} = \lambda\mathbf{x}$. Since Q is orthogonal, we have $\|Q\mathbf{x}\|_2 = \|\mathbf{x}\|_2$; so $\|\lambda\mathbf{x}\|_2 = \|\mathbf{x}\|_2$; so $|\lambda|\|\mathbf{x}\|_2 = \|\mathbf{x}\|_2$. Since $\mathbf{x} \neq \mathbf{0}$ and so $\|\mathbf{x}\|_2 \neq 0$, we must have $|\lambda| = 1$. Since $\lambda \in \mathbb{R}$, we must have either $\lambda = 1$ or $\lambda = -1$.

(b) Let $R \in \mathbb{R}^{n \times n}$ be an upper triangular matrix. What are the eigenvalues of R? Prove your claim.

SOLUTION. We claim that the eigenvalues of R are precisely its diagonal entries. We will prove this by induction on n. For $n = 1$, $R = [r_{11}]$ clearly has eigenvalue r_{11} correpsonding to the eigenvector 1. Suppose this is true for $n = k - 1$. For $n = k$, let

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & \cdots & r_{1k} \\ 0 & r_{22} & \cdots & \cdots & r_{2k} \\ 0 & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & r_{kk} \end{bmatrix} \in \mathbb{R}^{k \times k}.$$

小红书
小红书号:3189205503 qq

We will partition R into a 2×2 block matrix

$$R = \begin{bmatrix} R' & \mathbf{r} \\ \mathbf{0}^\top & r_{kk} \end{bmatrix}$$

where

$$R' = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1,k-1} \\ 0 & r_{22} & \cdots & r_{2,k-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{k-1,k-1} \end{bmatrix} \in \mathbb{R}^{(k-1)\times(k-1)}, \quad \mathbf{r} = \begin{bmatrix} r_{1k} \\ r_{2k} \\ \vdots \\ r_{k-1,k} \end{bmatrix} \in \mathbb{R}^{k-1}, \quad \mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{k-1}.$$

If $R'\mathbf{x} = \lambda\mathbf{x}$, then

$$R \begin{bmatrix} \mathbf{x} \\ 0 \end{bmatrix} = \begin{bmatrix} R' & \mathbf{r} \\ \mathbf{0}^\top & r_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 0 \end{bmatrix} = \begin{bmatrix} R'\mathbf{x} \\ 0 \end{bmatrix} = \begin{bmatrix} \lambda\mathbf{x} \\ 0 \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{x} \\ 0 \end{bmatrix}. \tag{3.1}$$

By the induction hypothesis, $r_{11}, \ldots, r_{k-1,k-1}$ are eigenvalues of R' and thus by (3.1), they are eigenvalues of R. If $r_{kk} \in \{r_{11}, \ldots, r_{k-1,k-1}\}$, then we are done. If $r_{kk} \notin \{r_{11}, \ldots, r_{k-1,k-1}\}$, then r_{kk} is not an eigenvalue of R' and so

$$(R' - r_{kk}I)\mathbf{y} \neq \mathbf{0}$$

for all $\mathbf{y} \neq \mathbf{0}$. So $\operatorname{nullsp}(R' - r_{kk}I) = \{\mathbf{0}\}$. So $R' - r_{kk}I$ is non-singular. Let $\mathbf{z} = -(R' - r_{kk}I)^{-1}\mathbf{r}$. Then $R'\mathbf{z} + \mathbf{r} = r_{kk}\mathbf{z}$. Therefore

$$R \begin{bmatrix} \mathbf{z} \\ 1 \end{bmatrix} = \begin{bmatrix} R' & \mathbf{r} \\ \mathbf{0}^\top & r_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{z} \\ 1 \end{bmatrix} = \begin{bmatrix} R'\mathbf{z} + \mathbf{r} \\ r_{kk} \end{bmatrix} = \begin{bmatrix} r_{kk}\mathbf{z} \\ r_{kk} \end{bmatrix} = r_{kk} \begin{bmatrix} \mathbf{z} \\ 1 \end{bmatrix}$$

and thus r_{kk} is an eigenvalue of R. In either case, $r_{11}, \ldots, r_{kk}$ are eigenvalues of R. Hence the result follows from induction.

(c) What is wrong with the following argument? Suppose we want to find the eigenvalues of $A \in \mathbb{R}^{n\times n}$ and we have found the QR-decomposition of A (say, via the Gram-Schmidt algorithm) to be

$$A = QR.$$

Now, if λ is an eigenvalue of Q and μ is an eigenvalue of R, then

$$A\mathbf{v} = QR\mathbf{v} = Q(\mu\mathbf{v}) = \mu Q\mathbf{v} = \mu\lambda\mathbf{v}$$

and so we have found $\mu\lambda$ to be an eigenvalue of A.

SOLUTION. The mistake here is to assume that $\mathbf{v}$ is a common eigenvector of Q and R corresponding to the eigenvalues λ and μ respectively. In general we should only have $Q\mathbf{x} = \lambda\mathbf{x}$ and $R\mathbf{y} = \mu\mathbf{y}$ — $\mathbf{x}$ and $\mathbf{y}$ need not be the same vector.

Remark: while the argument in (c) is incorrect, the QR-decomposition may indeed be used to compute eigenvalues and eigenvectors as we have discussed in the lectures; in fact, the QR-algorithm for computing eigenvalues was named one of the Top Ten Algorithms of the 20th Century: `http://math.berkeley.edu/~lekheng/courses/110/top10`.

小红书
小红书号:3189205503 qq